



AMRITA

VISHWA VIDYAPEETHAM

DEEMED TO BE UNIVERSITY

”

USER INTERFACE DESIGN

23CSE113

SUBMITTED BY	
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ROLL NO	AV.SC.U4CSE25111
YEAR/SEM/SECTION	1ST YEAR/ SEM-2/CSE-B
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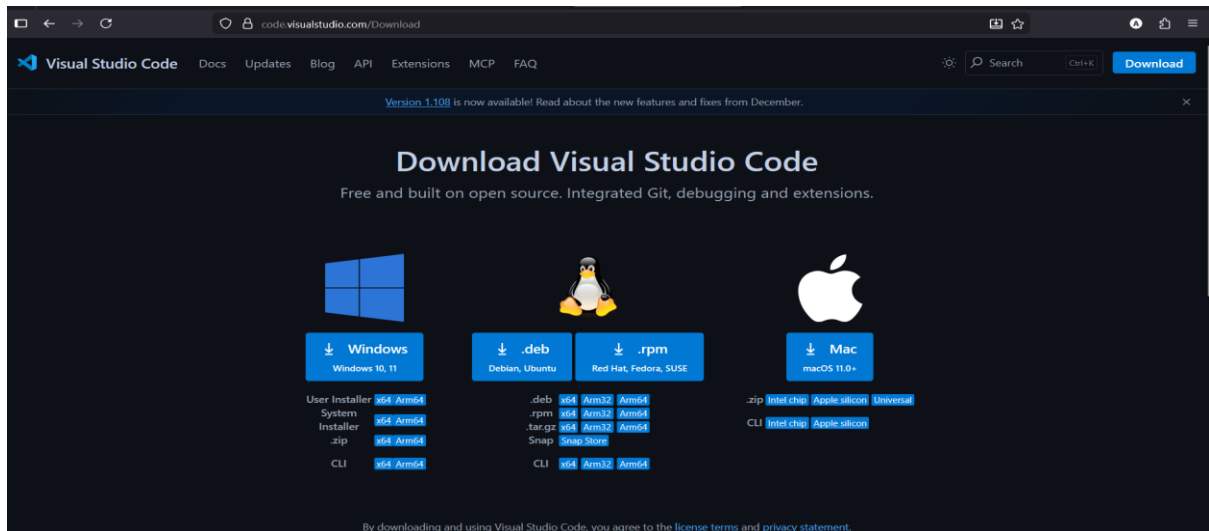
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LAB-1

1. AIM: Installing and setting up VS code

Procedure:

Step-1:



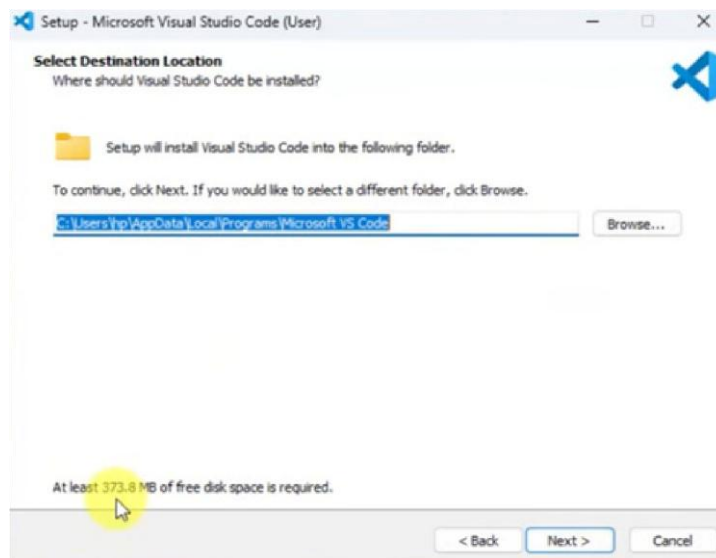
Go to Visualstudio.com and install VScode based on the operating system.

Step-2:



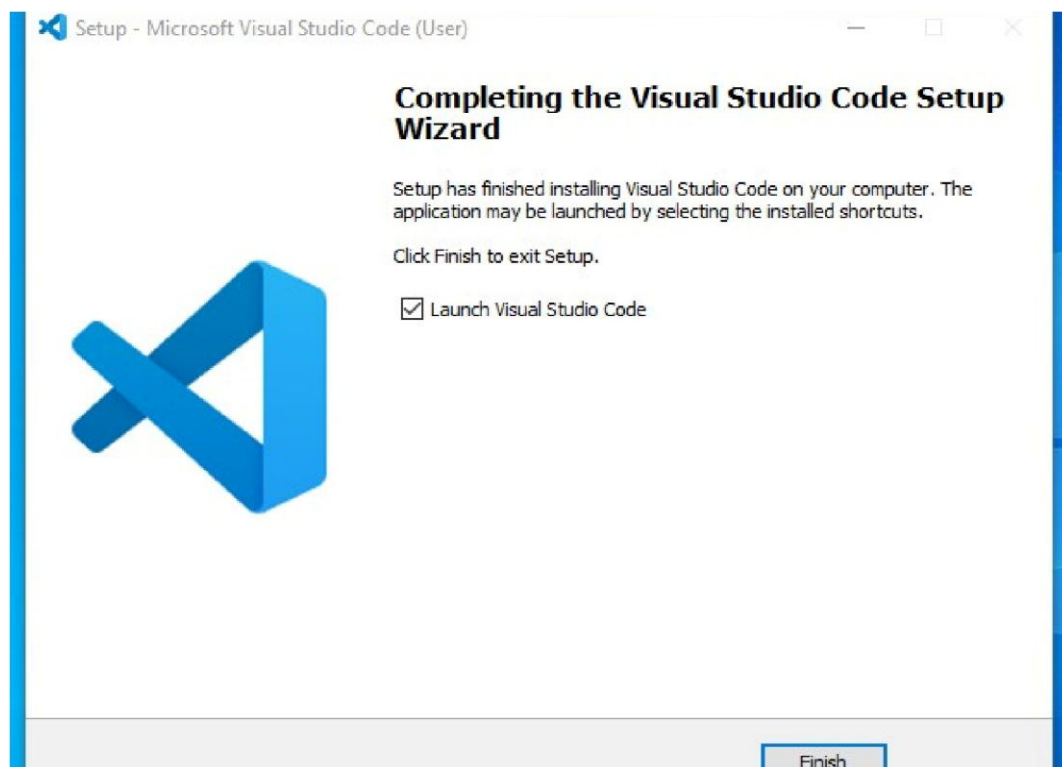
Accept the licence agreement and click on continue

Step-3:



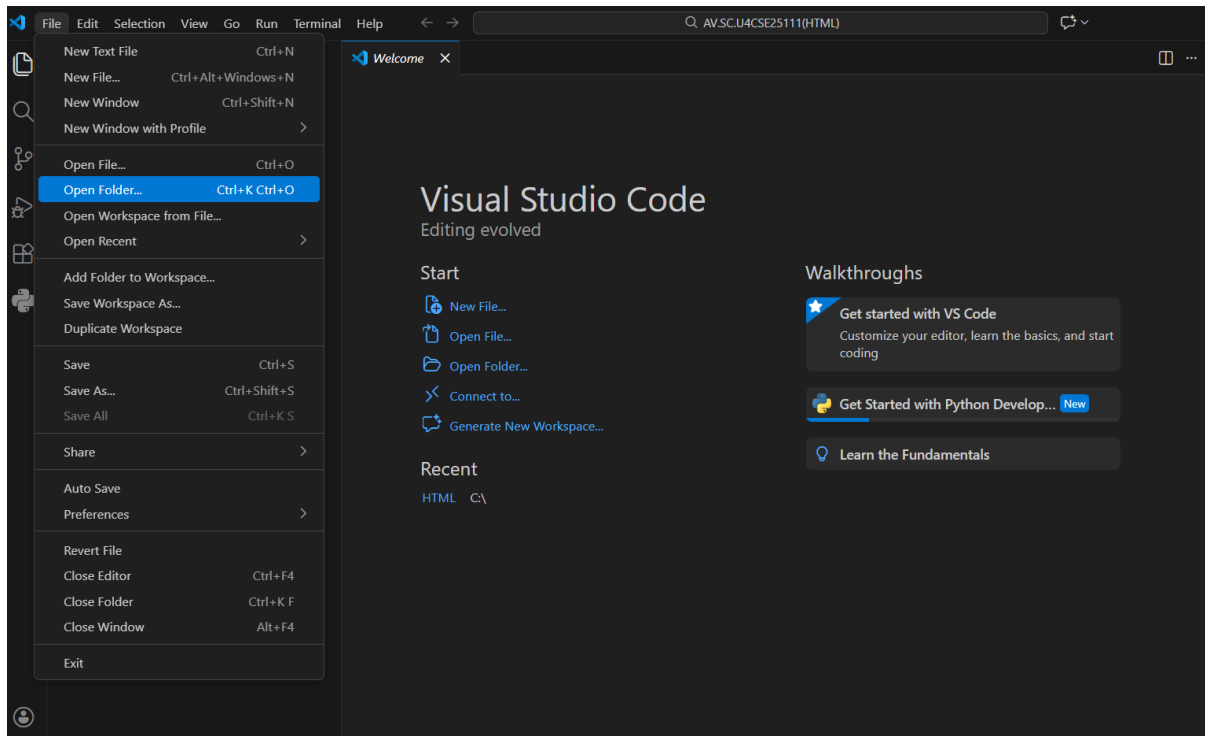
Select a folder to store VS Code

Step-4:



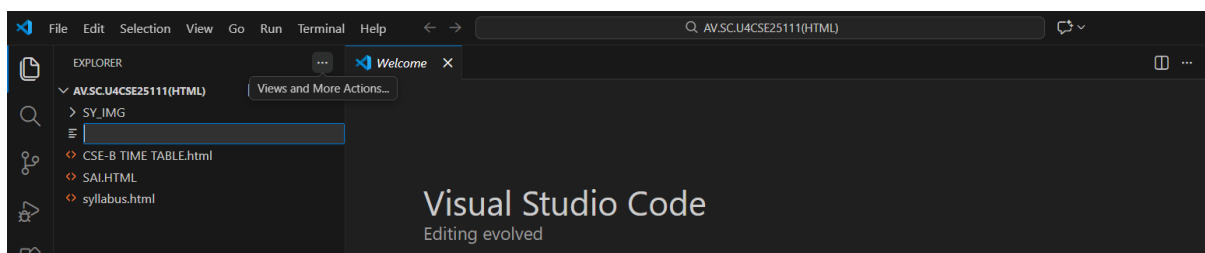
Click on finish to complete the installation

Step-5:



Click on file and click open folder and select the folder the folder you want to code on.

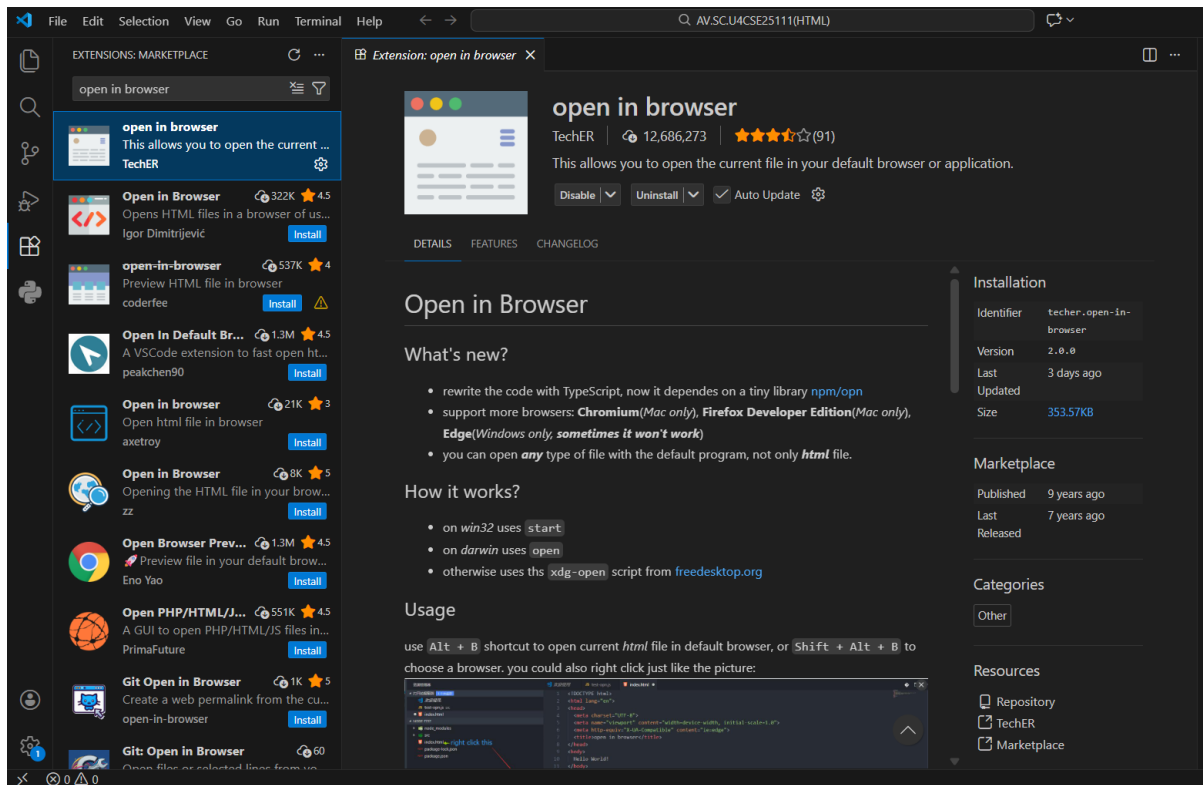
Step-6:



Click on new file and save it with the specific language suffix

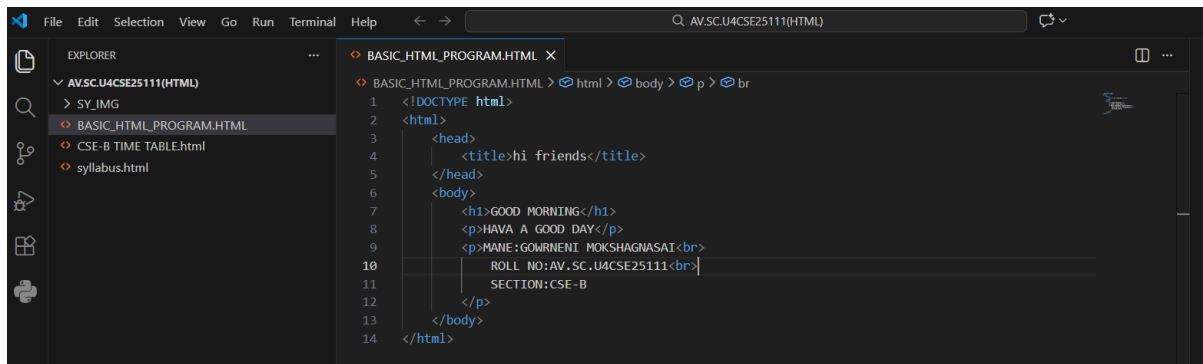
Eg: .html, .js, .python, etc

Step-7:



Download open in browser in vscode to run the HTML code.

Step-8:



To run the code right click the code and choose open in default browser or choose the browser you want to run the code

2. AIM: Structure of HTML

Program:

```
<!DOCTYPE html>

<html>

  <head>

    <title>HTML STRUCTURE</title>

  </head>

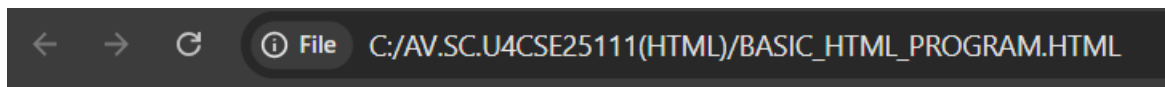
  <body>

    <p>NAME:GOWRNENI MOKSHAGNASAI<br>
    ROLL.NO:AV.SC.U4CSE25111<br> SECTION:CSE-B</p>

  </body>

</html>
```

Output:



GOOD MORNING

HAVA A GOOD DAY

MANE:GOWRNENI MOKSHAGNASAI
ROLL NO:AV.SC.U4CSE25111
SECTION:CSE-B

Explanation:

- HTML stands for hyper text markup language. It is used to create and structure the content of web pages.
- <!DOCTYPE html> tells the browser that this is HTML5 and it always goes at the top and it has no closing tag.
- <html></html> is the root element of the page. Everything must be inside this tag.
- <head></head> Contains information about the page <title>, <meta>, <link> and <script> are some of the common tags used in the head tag.

- `<title></title>` appears on the browser tab. Important for SEO (Search Engine Optimization).
- `<body></body>` Contains everything visible on the website like text, images, buttons, links, videos, etc. some common tags used in body are `<h1>`-`<h6>`, `<p>`, `<img>`, `<a>`, `<br>`, etc.

3. Aim: Basic Tags of HTML

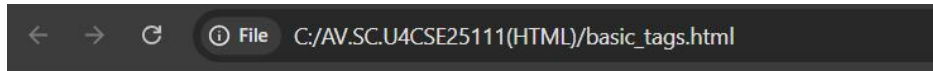
Program:

```

2  <html>
6  <body>
10 </body>
11 <h1>This is a h1 heading</h1>
12
13 <h2>This is a h2 heading</h2>
14
15 <h3>This is a h3 heading</h3>
16
17 <h4>This is a h4 heading</h4>
18
19 <h5>This is a h5 heading</h5>
20
21 <h6>This is a h6 heading</h6>
22 <br>
23 <p>This text formatting image</p>
24
25 <p style="background-color: yellow;">Image Of IronMan</p>
26
27 
28
29 <a href="https://www.amrita.edu/">This is a link to Amrita University Webpage</a>
30
31 </body>
32 </html>

```

Output:



MANE:GOWRNENI MOKSHAGNASAI
ROLL NO:AV.SC.U4CSE25111
SECTION:CSE-B

This is a h1 heading

This is a h2 heading

This is a h3 heading

This is a h4 heading

This is a h5 heading

This is a h6 heading

This text forming image

Image Of IronMan



[This is a link to Amrita University Webpage](#)

Explanation:

- `<h1>` - `<h6>` : Used for headings. `<h1>` is the largest and `<h6>` is the smallest. The font size decreases as we go from `<h1>` to `<h6>`. These are mainly used for headings.
- `<br>` (linebreak): Breaks the line and moves content to the next line. Empty tag (no closing tag).
- `<p>` (paragraph): In HTML, the element is used to define a paragraph. It's one of the most basic and commonly used tags in web development. Represents a block of text as a paragraph.
- `<img>` (image): image tag is used to add pictures in the webpage. To add pictures we need paste the path of the picture in the 'src' attribute of `<img>` tag. We can also use 'alt' attribute it is used to display a small text about the image if the image is not presented in the webpage.
- `<a>` `</a>`: The `<a>` is used to add links to the webpage. The link of the required webpage is pasted in the href attribute of the `<a>` tag and the text that is used to redirect to the page is given in between `<a>` and `</a>`.
- Style: style is used to customize the webpage. We can use style in many tags like `<h1>` to `<h6>`, `<p>`, `<img>`, etc. By using style we can add attributes like border, text color, text alignment, background color, font size, font style, etc.

LAB-2

2A.

AIM: To design a simple web page using HTML that demonstrates the use of background color, background image, heading styling, and preformatted text.

Program:

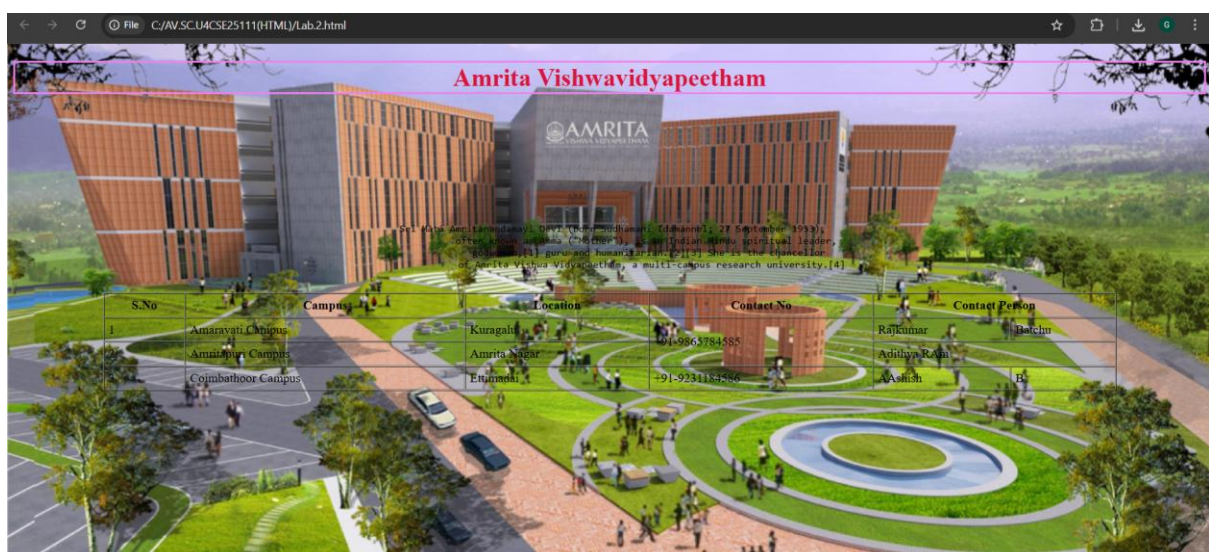
```
<!DOCTYPE html>
<html>
  <head>
    <title>Amrita</title>
  </head>
  <body style="background-image: url(amrita.png);background-repeat: no-
repeat;background-size: cover;">
    <center>
      <h1 style="color: crimson;border: 3px violet solid; text-align:center; ">Amrita
Vishwavidyapeetham</h1>
      <br>
      <br>
      <br>
      <br>
      <br>
      <br>
      <br>
      <pre style="text-align: center; color: deep teal;"> Sri Mātā Amritānandamayī Devi (born
Sudhamani Idamannel; 27 September 1953),
      often known as Amma ("Mother"), is an Indian Hindu spiritual leader,
      godwoman,[1] guru and humanitarian.[2][3] She is the chancellor
      of Amrita Vishwa Vidyapeetham, a multi-campus research university.[4]
    </pre>
    <table border="1" cellpadding="5" cellspacing="1" width="85%" align="center"
style="background-image:url(amritaamaravati.jpg);">
      <tr>
        <th>S.No</th>
        <th>Campus</th>
        <th>Location</th>
        <th>Contact No</th>
        <th colspan="2">Contact Person</th>
      </tr>
      <tr>
```

```

<td>1</td>
<td>Amaravati Campus</td>
<td>Kuragalu</td>
<td rowspan="2">+91-9865784585</td>
<td>Rajkumar</td>
<td>Batchu</td>
</tr>
<tr>
<td>2</td>
<td>Amritapuri Campus</td>
<td>Amrita Nagar</td>
<td colspan="2">Adithya RAm</td>
</tr>
<tr>
<td>3</td>
<td>Coimbatheoor Campus</td>
<td>Ettimadai</td>
<td>+91-9231184586</td>
<td>AAshish</td>
<td>B</td>
</tr>
</table>
</center>
</body>
</html>

```

Output:



Explanation:

The HTML program uses several important tags and attributes to structure and style the web page for Amrita Vishwa Vidyapeetham. The `<!DOCTYPE html>` tag defines the HTML5 document type, while `<html>` is the root element containing the whole page. Inside it, `<head>` stores metadata and the `<title>` sets the browser tab name. The `<body>` tag holds all visible content and uses the `style` attribute to apply CSS properties like `background-image`, `background-repeat`, and `background-size`. Text content is organized using `<h1>` for headings, `<br>` for line breaks, and `<pre>` for preformatted text that preserves spacing. Layout is centered using the `<center>` tag (though it is obsolete in modern HTML). Tabular data is created with `<table>`, with rows defined by `<tr>`, header cells by `<th>`, and data cells by `<td>`. Table formatting attributes include `border`, `cellpadding`, `cellspacing`, `width`, and `align`, while cell-merging is achieved using `rowspan` and `colspan`. Inline styling is applied through the `style` attribute to control color, border, and text alignment.

2B.

AIM:

To create a static webpage using only HTML that displays basic information about Amrita Vishwa Vidyapeetham, including a heading, descriptive text, background styling through HTML attributes, and a table showing the list of campuses in different states.

Program:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Amrita vishyavidyapeetham</title>
  </head>
  <body style="background-image: url(https://c8.alamy.com/comp/HY6DWB/main-
building-of-the-amrita-university-campus-in-ettimadai-coimbatore-HY6DWB.jpg)
;background-repeat: no-repeat;background-size:cover" >
    <h1>Amrita vishyavidyapeetham</h1>
    <p>Amrita Vishwa Vidyapeetham is a well-known private university in India,
founded by Mata Amritanandamayi Devi (Amma). It has multiple campuses
across the country, including Coimbatore and Kerala, offering a wide
range of undergraduate, postgraduate, and doctoral programs. The university
is recognized for its strong emphasis on research, innovation, and value-based
education. It focuses not only on academic excellence but also on ethical and
```


social responsibility. Amrita collaborates with many international institutions, enhancing global learning opportunities for students. It is consistently ranked among the top universities in India and provides modern infrastructure, advanced laboratories, and digital learning facilities. The institution aims to develop skilled professionals who contribute positively to society.</p>

```
<center><table border="1" cellspacing="3" cellpadding="5">
  <tr>
    <th>s.no</th>
    <th>campus-state</th>
  </tr>
  <tr>
    <td>1</td>
    <td>Coimbatore</td>
  </tr>
  <tr>
    <td>2</td>
    <td>Amaravati</td>
  </tr>
  <tr>
    <td>3</td>
    <td>Chennai</td>
  </tr>
  <tr>
    <td>4</td>
    <td>Bengaluru</td>
  </tr>
  <tr>
    <td>5</td>
    <td>Faridabad</td>
  </tr>
  <tr>
    <td>6</td>
    <td>kochi</td>
  </tr>
  <tr>
    <td>7</td>
    <td>mysuru</td>
  </tr>
</table>
</center>

</body>
</html>
```

OUTPUT:



EXPLANATION:

The <html> document is used to create a webpage about Amrita Vishwa Vidyapeetham, where the <head> section contains the title of the page and styling using CSS to improve the design. The <body> section displays the content and includes a background image that covers the entire screen using background-size: cover. The <h1> tag is used to show the main heading, and the <p> tag provides a detailed description of the university. To make the text readable over the background image, a semi-transparent background color is applied using CSS. A <table> is placed at the center of the page to list different campuses, where <th> represents the table headings and <td> represents the data cells. Additional styling like padding, borders, and colors is used to make the table neat and visually appealing, resulting in a well-structured and attractive webpage.

2C.

AIM: To design a restaurants menu webpage using HTML and displaying menu items in a table format with proper layout and visual styling.

PROGRAM:

```
<!DOCTYPE html>
<html>
  <head>
    <title>
      Restaurant Menu
    </title>
  </head>
  <body style="font-family: 'Gill Sans', 'Gill Sans MT', Calibri, 'Trebuchet MS', sans-serif;
background-image: url(https://plus.unsplash.com/premium_photo-1661883237884-
```

```

263e8de8869b?fm=jpg&q=60&w=3000&auto=format&fit=crop&ixlib=rb-
4.1.0&ixid=M3wxMjA3fDB8MHxzZWfyY2h8Mnx8cmVzdGF1cmFudHxlbmwHwwfHx
8MA%3D%3D);font-weight: 200;background-color:yellow; text-align: center; ">
<p style="font-weight: 200;background-color:yellow; text-align: center; ">Welcome</p>
<table border="1" cellspacing="3" cellpadding="5"
style="width: 600px; margin: 100px auto; padding:20px;
background-color: rgba(255, 255, 255, 0.2);
backdrop-filter: blur(10px);
border-radius: 10px; color: black;">
<tr>
<th>Item</th>
<th>Description</th>
<th>Price</th>
</tr>
<tr>
<td colspan="3">Deserts</td>
</tr>
<tr>
<td>Veg spring roll</td>
<td>Crispy rolls with veggies-6pic</td>
<td>120</td>
</tr>
<tr>
<td>KFC</td>
<td>Spicy grilled wing 8-pic</td>
<td>240</td>
</tr>
<tr>
<td colspan="3">main course</td>
</tr>
<tr>
<td>Special Mushroom Biryani</td>
<td>Creamy Mushroom curry</td>
<td>210</td>
</tr>
<tr>
<td>Chicken Biryani</td>
<td>Vijayawada style</td>
<td>240</td>
</tr>
<tr>
<td colspan="3">Starters</td>
</tr>
<tr>
<td>Ice-cream</td>

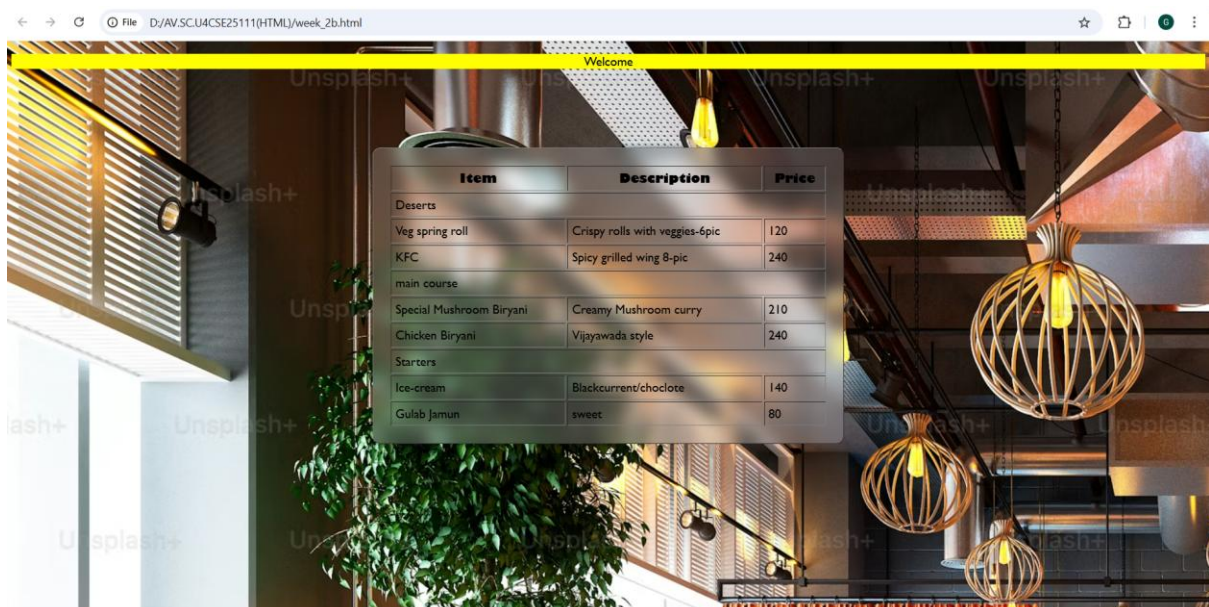
```

```

        <td>Blackcurrent/choclote</td>
        <td>140</td>
    </tr>
    <tr>
        <td>Gulab Jamun</td>
        <td>sweet</td>
        <td>80</td>
    </tr>
</table>
</body>
</html>

```

OUTPUT:



EXPLANATION:

The HTML code creates a simple restaurant menu webpage where the `<html>` structure contains a `<head>` section with the title “Restaurant Menu” and a `<body>` section that displays the content. The `<body>` is styled with a background image and font settings to give a visually appealing look, and the text is centered on the page. A `<p>` tag is used to display a welcome message at the top. The main content is presented using a `<table>`, which is styled with width, margin, padding, and a semi-transparent background using `rgba` and `backdrop-filter` to create a glass effect. Inside the table, `<th>` tags define the headings (Item, Description, Price), while `<td>` tags contain the menu details. The `colspan="3"` attribute is used to create section headers like desserts, main course, and starters across all columns. Overall, the code combines HTML structure and inline CSS styling to create an organized and attractive restaurant menu layout.

LAB-3

AIM: To design and develop an academic timetable using HTML tables by applying row span, column span, and styling for proper structure and presentation.

Program:

```
<!DOCTYPE html>
<html>
  <head>
    <style>
      table, th, td {
        border: 1px solid black;
        border-collapse: collapse;
      }
    </style>
  </head>
  <body>
    <h1 style="text-align: center;">CSE-B TIME TABLE</h1>
    <table style="width:100%"style="text-align: center;">
      <th colspan="12">Master Time Table CSE II Semester B Section</th>
      <tr>
        <td colspan="4">Dept. CSE</td>
        <td rowspan="3" colspan="8">Class Room Venu:511</td>
      </tr>
      <tr>
        <td colspan="4"> Class: B. Tech. CSE-B</td>
      </tr>
      <tr>
        <td colspan="4">Semester-II</td>
      </tr>
      <tr>
        <td colspan="4"><p></p>
          Class Advisors:<br>
          1. Name: Dr. Kurmala Gowri Raghavendra Narayan<br>
          Email : kgr\_narayan@av.amrita.edu<br>
          Mobile Number : +91-9705409393</p></td>
        <td colspan="8"><p>
          2. Name: Dr. Ankit Sachan<br>
          Email : s\_ankit@av.amrita.edu<br>
          Mobile Number :+91-8381800129
        </p>
      </tr>
```

```

        </td>
    </tr>
    <tr>
        <td rowspan="2">TIME/DAY</td>
        <td style="text-align: center;">Slot 1</td>
        <td style="text-align: center;">Slot 2</td>
        <td style="text-align: center;" rowspan="2">10.35am-10.45am</td>
        <td style="text-align: center;">Slot 3</td>
        <td style="text-align: center;">Slot 4</td>
        <td style="text-align: center;">Slot 5</td>
        <td style="text-align: center;" rowspan="2">01.15pm-02.05pm</td>
        <td style="text-align: center;">Slot 6</td>
        <td style="text-align: center;" rowspan="2"> 3.00pm-3.10pm</td>
        <td style="text-align: center;">Slot 7</td>
        <td style="text-align: center;">Slot 8</td>
    </tr>
    <tr>
        <td style="text-align: center;"> 08.55-09.45am</td>
        <td style="text-align: center;"> 09.45-10.35am</td>
        <td style="text-align: center;"> 10.45-11.35am</td>
        <td style="text-align: center;"> 11.35am-12.25pm</td>
        <td style="text-align: center;"> 12.25am-01.15pm</td>
        <td style="text-align: center;"> 02.10-03.00pm</td>
        <td style="text-align: center;"> 03.10-04.00pm</td>
        <td style="text-align: center;"> 04.00-04.50pm</td>
    </tr>
    <tr>
        <td>Monday</td>
        <td colspan="2">23MAT117 Linear Algebra LAB(BYOD)</td>
        <td rowspan="5" style="writing-mode: vertical-lr" ; text-align: center;
bgcolor="yellow"><pre style="font-size: xx-large;">    Short Break    </pre></td>
        <td>22ADM111<br> Glimpses of<br> Glorious India</td>
        <td>23PHY115 Modern<br> Physics</td>
        <td style="text-align: center;"> LIBRARY</td>
        <td rowspan="5" style="writing-mode: vertical-lr; text-align: center;"
bgcolor="yellow">Lunch Break</td>
        <td style="text-align: center;">23MEE115 <br>Manufacturing Practice
<br>LAB</td>
        <td rowspan="5" style="writing-mode: vertical-lr" ; text-align: center;
bgcolor="yellow"><pre style="font-size: xx-large;">    Short Break    </pre></td>
        <td colspan="2">23MEE115 Manufacturing Practice LAB</td>
    </tr>

```



```

<tr>
  <td>Tuesday</td>
  <td> 23MAT116 Discrete <br>Mathematics</td>
  <td>23MAT117 Linear <br>Algebra</td>
  <td>23PHY115 <br>Modern Physics</td>
  <td>23CSE111Object <br> Oriented Programming</td>
  <td>23CSE113 User <br>Interface Design</td>
  <td>22ADM111Glimpses of Glorious India(P)</td>
  <td style="text-align: center;" colspan="2">Library/Sport</td>
</tr>
<tr>
  <td>Wednesday</td>
  <td style="text-align: center;" colspan="2">23CSE111Object Oriented
Programming_lab2</td>
  <td>23MAT116 <br>Discrete <br>Mathematics</td>
  <td>23CSE113 User <br>Interface Design</td>
  <td>22ADM111Glimpses <br>of Glorious India</td>
  <td style="text-align: center;" colspan="2">Library/Sports</td>
  <td style="text-align: center;" colspan="2">Library/Sport</td>
</tr>
<tr>
  <td>Thursday</td>
  <td style="text-align: center;" colspan="2">23MAT117 Linear Algebra</td>
  <td> 23CSE111 Object <br>Oriented Programming</td>
  <td>23MAT116 <br>Discrete<br> Mathematics</td>
  <td style="text-align: center;" bgcolor="red">COUNSELLING</td>
  <td style="text-align: center;" colspan="2">LIBRARY </td>
  <td style="text-align: center;" colspan="2">23PHY115 Modern <br>Physics(T)</td>
  <td style="text-align: center;" colspan="2">Library/Sport</td>
</tr>
<tr>
  <td>Friday</td>
  <td style="text-align: center;" colspan="2">23CSE113 User Interface Design Lab
BYOD</td>
  <td> 23MAT117 <br>Linear Algebra</td>
  <td> 23CSE111 Object <br>Oriented Programming</td>
  <td style="text-align: center;" colspan="2">Library/Sports</td>
  <td style="text-align: center;" bgcolor="red">COUNSELLING</td>
  <td style="text-align: center;" colspan="2">23MAT116 Discrete
Mathematics_lab_lab1</td>
</tr>

```

```

<tr>

  <td style="text-align: center;">Course Code</td>
  <td style="text-align: center;" colspan="2">Course Name</td>
  <td></td>
  <td style="text-align: center;">L-T-P</td>
  <td>credits</td>
  <td colspan="3" style="text-align: center;"> Faculty Name</td>
  <td></td>
  <td style="text-align: center;" colspan="2"> Assisting Faculty</td>
</tr>

<tr>
  <td style="text-align: center;">23MAT116</td>
  <td style="text-align: center;" colspan="2">Discrete Mathematics</td>
  <td></td>
  <td style="text-align: center;">3-0-2</td>
  <td style="text-align: center;">4</td>
  <td style="text-align: center;" colspan="3"> Dr.Srivivasa Rao Thota</td>
  <td></td>
  <td style="text-align: center;" colspan="2"></td>
</tr>

<tr>
  <td style="text-align: center;">23CSE111</td>
  <td style="text-align: center;" colspan="2">Object Oriented Programming</td>
  <td></td>
  <td style="text-align: center;">3-0-2</td>
  <td style="text-align: center;">4</td>
  <td style="text-align: center;" colspan="3">Dr. Balaji TK</td>
  <td></td>
  <td style="text-align: center;" colspan="2"></td>
</tr>

<tr>
  <td style="text-align: center;">23PHY115</td>
  <td style="text-align: center;" colspan="2">Modern Physics</td>
  <td></td>
  <td style="text-align: center;">2-1-0</td>
  <td style="text-align: center;">3</td>
  <td style="text-align: center;" colspan="3">Dr. C R Kesavalu</td>
  <td></td>
  <td style="text-align: center;" colspan="2"></td>

```



```

<td style="text-align: center;" colspan="2"></td>
</tr>

<tr>
<td style="text-align: center;">23CSE113</td>
<td style="text-align: center;" colspan="2">User Interface Design</td>
<td></td>
<td style="text-align: center;">2-0-2</td>
<td style="text-align: center;">3</td>
<td style="text-align: center;" colspan="3"> Dr.Rajkumar Batchu</td>
<td></td>
<td style="text-align: center;" colspan="2"> Mr. Barge Bharadwaj</td>
</tr>

<tr>
<td style="text-align: center;">23MEE115</td>
<td style="text-align: center;" colspan="2">Manufacturing Practice</td>
<td></td>
<td style="text-align: center;">0-0-3</td>
<td style="text-align: center;">1</td>
<td style="text-align: center;" colspan="3"> Dr. Sathies S.</td>
<td></td>
<td style="text-align: center;" colspan="2"></td>
</tr>

<tr>
<td style="text-align: center;">22ADM111</td>
<td style="text-align: center;" colspan="2">Glimpses of Glorious India</td>
<td></td>
<td style="text-align: center;">2-0-1</td>
<td style="text-align: center;">2</td>
<td style="text-align: center;" colspan="3"> Dr. Siva Panuganti</td>
<td></td>
<td style="text-align: center;" colspan="2"></td>
</tr>

<tr>
<td></td>
<td colspan="2" style="text-align: center;">Total (15L + 1Tut +12P =28 hrs)</td>
<td></td>
<td style="text-align: center;">15-1-12</td>
<td style="text-align: center;">21</td>

```

```

</tr>
</table>
</body>
</html>

```

Output:

<

Explanation:

This program is an HTML webpage that displays a class timetable using a table layout. It uses basic HTML tags like `<table>`, `<tr>`, `<td>`, and `<th>` to organize the schedule into rows and columns. CSS styling is added in the head section to give borders and a clean look to the table. The program uses `colspan` and `rowspan` to merge cells for headings, breaks, and lab periods. It includes the timetable slots, subject names, break times, faculty details, and course information, all arranged in a structured format so it appears like an official timetable when opened in a browser.

AIM: To create a comprehensive webpage that includes an academic timetable along with syllabus representation using images and structured HTML elements.

Program:

```
<!DOCTYPE html>
<html>
  <head>
    <title>SYLLABUS Of 2nd Semester</title>
    <style>
      table, th, td {
        border: 1px solid black;
        border-collapse: collapse;
      }
    </style>
  </head>
  <body style="background-color: orange;">
    <h1 style="text-align: center;">SYLLABUS</h1>
    <h2>Discrete Mathematics</h2>
    
    <h2>Linear Algebra</h2>
    
    <h2>Object Oriented Programming Language</h2>
    
    <h2>Modern Physics</h2>
    
    <h2>User Interface Design</h2>
    
    <h2>Manufacturing Practice</h2>
    
    <h2>Glimpses of Glorious India</h2>
    

    <h1 style="text-align: center;">CSE-B TIME TABLE</h1>
    <table style="width:100%" style="text-align: center;">
      <th colspan="12">Master Time Table CSE II Semester B Section</th>
      <tr>
        <td colspan="4">Dept. CSE</td>
        <td rowspan="3" colspan="8">Class Room Venu:511</td>
      </tr>
      <tr>
        <td colspan="4">Class: B. Tech. CSE-B</td>
      </tr>
```

```

<tr>
  <td colspan="4">Semester-II</td>
</tr>
<tr>
  <td colspan="4"><p></p>
    Class Advisors:<br>
    1. Name: Dr. Kurmala Gowri Raghavendra Narayan<br>
    Email : kgr\_narayan@av.amrita.edu<br>
    Mobile Number : +91-9705409393</p></td>
    <td colspan="8"><p>
      2. Name: Dr. Ankit Sachan<br>
      Email : s\_ankit@av.amrita.edu<br>
      Mobile Number :+91-8381800129
    </p>
  </td>
</tr>
<tr>
  <td rowspan="2">TIME/DAY</td>
  <td style="text-align: center;">Slot 1</td>
  <td style="text-align: center;">Slot 2</td>
  <td style="text-align: center;" rowspan="2">10.35am-10.45am</td>
  <td style="text-align: center;">Slot 3</td>
  <td style="text-align: center;">Slot 4</td>
  <td style="text-align: center;">Slot 5</td>
  <td style="text-align: center;" rowspan="2">01.15pm-02.05pm</td>
  <td style="text-align: center;">Slot 6</td>
  <td style="text-align: center;" rowspan="2"> 3.00pm-3.10pm</td>
  <td style="text-align: center;">Slot 7</td>
  <td style="text-align: center;">Slot 8</td>
</tr>
<tr>
  <td style="text-align: center;"> 08.55-09.45am</td>
  <td style="text-align: center;"> 09.45-10.35am</td>
  <td style="text-align: center;"> 10.45-11.35am</td>
  <td style="text-align: center;"> 11.35am-12.25pm</td>
  <td style="text-align: center;"> 12.25am-01.15pm</td>
  <td style="text-align: center;"> 02.10-03.00pm</td>
  <td style="text-align: center;"> 03.10-04.00pm</td>
  <td style="text-align: center;"> 04.00-04.50pm</td>
</tr>
<tr>
  <td>Monday</td>

```

```

<td colspan="2">23MAT117 Linear Algebra LAB(BYOD)</td>
<td rowspan="5" style="writing-mode: vertical-lr" ; text-align: center;
bgcolor="yellow"><pre style="font-size: xx-large;">    Short Break    </pre></td>
<td>22ADM111<br> Glimpses of<br> Glorious India</td>
<td>23PHY115 Modern <br> Physics</td>
<td style="text-align: center;"> LIBRARY</td>
<td rowspan="5" style="writing-mode: vertical-lr; text-align: center;"
bgcolor="yellow">Lunch Break</td>
<td style="text-align: center;">23MEE115 <br>Manufacturing Practice
<br>LAB</td>
<td rowspan="5" style="writing-mode: vertical-lr" ; text-align: center;
bgcolor="yellow"><pre style="font-size: xx-large;">    Short Break    </pre></td>
<td colspan="2">23MEE115 Manufacturing Practice LAB</td>
</tr>
<tr>
<td>Tuesday</td>
<td> 23MAT116 Discrete <br>Mathematics</td>
<td>23MAT117 Linear <br>Algebra</td>
<td>23PHY115 <br>Modern Physics</td>
<td>23CSE111Object <br> Oriented Programming</td>
<td>23CSE113 User <br>Interface Design</td>
<td>22ADM111Glimpses of Glorious India(P)</td>
<td style="text-align: center;" colspan="2">Library/Sport</td>
</tr>
<tr>
<td>Wednesday</td>
<td style="text-align: center;" colspan="2">23CSE111Object Oriented
Programming_lab2</td>
<td>23MAT116 <br>Discrete <br>Mathematics</td>
<td>23CSE113 User <br>Interface Design</td>
<td>22ADM111Glimpses <br>of Glorious India</td>
<td style="text-align: center;">Library/Sports</td>
<td style="text-align: center;" colspan="2">Library/Sport</td>
</tr>
<tr>
<td>Thursday</td>
<td style="text-align: center;">23MAT117 Linear Algebra</td>
<td> 23CSE111 Object <br>Oriented Programming</td>
<td>23MAT116 <br>Discrete<br> Mathematics</td>
<td style="text-align: center;" bgcolor="red">COUNSELLING</td>
<td style="text-align: center;">LIBRARY </td>
<td style="text-align: center;">23PHY115 Modern <br>Physics(T)</td>

```

```

        <td style="text-align: center;" colspan="2">Library/Sport</td>
    </tr>
    <tr>
        <td>Friday</td>
        <td style="text-align: center;" colspan="2">23CSE113 User Interface Design Lab
BYOD</td>
        <td> 23MAT117 <br>Linear Algebra</td>
        <td> 23CSE111 Object <br>Oriented Programming</td>
        <td style="text-align: center;">Library/Sports</td>
        <td style="text-align: center;" bgcolor="red">COUNSELLING</td>
        <td style="text-align: center;" colspan="2">23MAT116 Discrete
Mathematics_lab_lab1</td>
    </tr>

    <tr>

        <td style="text-align: center;">Course Code</td>
        <td style="text-align: center;" colspan="2">Course Name</td>
        <td></td>
        <td style="text-align: center;">L-T-P</td>
        <td>credits</td>
        <td colspan="3" style="text-align: center;"> Faculty Name</td>
        <td></td>
        <td style="text-align: center;" colspan="2"> Assisting Faculty</td>
    </tr>

    <tr>
        <td style="text-align: center;">23MAT116</td>
        <td style="text-align: center;" colspan="2">Discrete Mathematics</td>
        <td></td>
        <td style="text-align: center;">3-0-2</td>
        <td style="text-align: center;">4</td>
        <td style="text-align: center;" colspan="3"> Dr.Srivivasa Rao Thota</td>
        <td></td>
        <td style="text-align: center;" colspan="2"></td>
    </tr>

    <tr>
        <td style="text-align: center;">23CSE111</td>
        <td style="text-align: center;" colspan="2">Object Oriented Programming</td>
        <td></td>
        <td style="text-align: center;">3-0-2</td>

```

```

<td style="text-align: center;">4</td>
<td style="text-align: center;" colspan="3">Dr. Balaji TK</td>
<td></td>
<td style="text-align: center;" colspan="2"></td>
</tr>

<tr>
<td style="text-align: center;">23PHY115</td>
<td style="text-align: center;" colspan="2">Modern Physics</td>
<td></td>
<td style="text-align: center;">2-1-0</td>
<td style="text-align: center;">3</td>
<td style="text-align: center;" colspan="3">Dr. C R Kesavalu</td>
<td></td>
<td style="text-align: center;" colspan="2"></td>
</tr>

<tr>
<td style="text-align: center;">23CSE113</td>
<td style="text-align: center;" colspan="2">User Interface Design</td>
<td></td>
<td style="text-align: center;">2-0-2</td>
<td style="text-align: center;">3</td>
<td style="text-align: center;" colspan="3"> Dr.Rajkumar Batchu</td>
<td></td>
<td style="text-align: center;" colspan="2"> Mr. Barge Bharadwaj</td>
</tr>

<tr>
<td style="text-align: center;">23MEE115</td>
<td style="text-align: center;" colspan="2">Manufacturing Practice</td>
<td></td>
<td style="text-align: center;">0-0-3</td>
<td style="text-align: center;">1</td>
<td style="text-align: center;" colspan="3"> Dr. Sathies S.</td>
<td></td>
<td style="text-align: center;" colspan="2"></td>
</tr>

<tr>
<td style="text-align: center;">22ADM111</td>
<td style="text-align: center;" colspan="2">Glimpses of Glorious India</td>

```

```

<td></td>
<td style="text-align: center;">2-0-1</td>
<td style="text-align: center;">2</td>
<td style="text-align: center;" colspan="3"> Dr. Siva Panuganti</td>
<td></td>
<td style="text-align: center;" colspan="2"></td>
</tr>

<tr>
<td></td>
<td colspan="2" style="text-align: center;">Total (15L + 1Tut +12P =28 hrs)</td>
<td></td>
<td style="text-align: center;">15-1-12</td>
<td style="text-align: center;">21</td>
</tr>
</table>
</body>
</html>

```

Output:

The screenshot shows a web browser window with the address bar displaying 'C:/AV.SC.U4CSE2511(HTML)/syllabus.html'. The page has an orange header with the title 'SYLLABUS'. Below the header, the title 'Discrete Mathematics' is shown. The main content area is white and contains a 'Syllabus' section with three units: Unit I (Logic, Mathematical Reasoning and Counting), Unit II (Relations and Their Properties, Advanced Counting Techniques and Relations), and Unit III (Graph Theory). A 'Lab Practice Problems' section is also present at the bottom.

Linear Algebra

Syllabus

Unit I

Vector Spaces: Vector spaces – Sub spaces – Linear independence – Basis – Dimension; Inner Product Spaces: Inner products – Orthogonality – Orthogonal basis – Gram Schmidt Process – Change of basis – Orthogonal complements – Projection on subspace – Least Square Principle. QR- Decomposition.

Unit II

Functions of severable variables: Functions, limit and continuity, Partial differentiations, total derivatives, differentiation of implicit functions and transformation of coordinates by Jacobian. Taylor's series for two variables. Vector Differentiation: Vector and Scalar Functions, Derivatives, Curves, Tangents, Arc Length, Curves in Mechanics, Velocity and Acceleration, Gradient of a Scalar Field, Directional Derivative, Divergence of a Vector Field, Curl of a Vector Field.

Unit III

Eigen values and Eigen vectors: Eigen Values and Eigen Vectors, Diagonalization, Orthogonal Diagonalization, Quadratic Forms, Diagonalizing Quadratic Forms, Conic Sections. Similarity of linear transformations – Diagonalization and its applications – Jordan form and rational canonical form. Case Studies: Applications on least square and image transformations.

Object Oriented Programming Language

Syllabus

Unit I

Structured to Object Oriented Approach by Examples – Object Oriented languages – Properties of Object Oriented system – UML and Object Oriented Software Development – Use case diagrams and documents as a functional model – Identifying Objects and classes – Representation of Objects and its state by Object Diagram – Simple Class using class diagram – Encapsulation – Data Hiding – Reading and Writing Objects – Class Level and Instance Level Attributes and Methods- JIVE environment for debugging.

Unit II

Aggregation and Composition using Class Diagram – Generalization using Class Diagram – Inheritance – Constructor and Over Riding – Visibility – Attribute – Parameter – Package – Local and Global – Polymorphism – Overloading – Abstract Classes and Interfaces.

Unit III

Exception Handling – Inner Classes – Wrapper classes – String – and String Builder classes – Number – Math – Random – Array methods – File Streams – Serialization – Generics – Collection framework – Comparator and Comparable – Vector and ArrayList – Iterator and Iterable.

Modern Physics

Syllabus

Unit I

Origin of quantum theory of radiation: Black body radiation, photo-electric effect, Compton Effect – pair production and annihilation, De-Broglie hypothesis, description of waves and wave packets, group velocities. Evidence for wave nature of particles: Davission-Germer experiment, Heisenberg uncertainty principle.

Unit II

Atomic structure: Historical Development of atomic structures: Thomson's Model, Rutherford's Model: Scattering formula and its predictions, Atomic spectra – Bohr's Model, Sommerfield's Model, The correspondence principle, nuclear motion, and atomic excitation, Application: Lasers.

Unit III

Quantum mechanics: Wave function, Probability density, expectation values – Schrodinger equation – time dependent and independent, Linearity and superposition, expectation values, operators, Eigen functions and Eigen values

Unit IV

Application of 1D Schrodinger Wave equation: Free particle, Particle in a box, Finite potential well, Tunnel effect, Harmonic oscillator.

Unit V

Intro to Quantum computing- Q bits- II Quantum correlations: Bell inequalities and entanglement, Schmidt decomposition, super dense coding, teleportation. Module

User Interface Design

Syllabus

Unit I

Introduction to Internet and Web design – Working of Web – Roles of Front-end, Back-end and Full stack development – Web Server – Internet protocols – Web Hosting – HTML – HTML Tags – Create a simple Web Page – Marking up Text – Adding Links – Adding Images – SVG – Table Markup – Embedded Media – Web Based Forms – HTML Forms – CSS for Presentation – Basic Style Sheet – A CSS Style Primer – Using Style Classes – Using Style IDs – Formatting Text – Advanced Typography with CSS3 – Working with Margins, Padding, Alignment, and Floating. Responsive web design, Introduction to version control systems.

Unit II

CSS Box Model and Positioning – Creating Layouts Using Modern CSS – Backgrounds and Borders – CSS Transformations and Transitions – Animating with CSS and the Canvas – Dynamic Web Pages – Web Scripting – JavaScript programming for Web Applications – Including JavaScript in HTML – HTML5 Form Controls and Validation – Document Object Model – DOM manipulation and events – Event Handlers

Unit III

Overview of UI Frameworks – User Interface Design Concepts in Web Development – Accessibility – Introduction to Electron – Overview of Electron – Setting up an Electron project – Creating a basic desktop application with Electron – Object communication – Introduction to other popular frameworks and libraries (e.g., JavaFX, WPF, GTK).

Manufacturing Practice

Syllabus

Syllabus

1. Additive Manufacturing Laboratory –12 hours

Introduction to digital manufacturing. Introduction to Additive Manufacturing – types – additive manufacturing applications – Materials for 3D printing, CAD Modelling for Additive manufacturing, Slicing and STL file generation- G code generation – 3D printing of simple geometries.

2. Mechanical Engineering Laboratory –12 hours

Study of tools and equipment used for basic manufacturing processes.
Manual arc welding practice for making Butt and Lap joints – Soldering Practice
Introduction to Machine Tools and Machining Processes

3. Automation lab –12 hours

Design, simulate and test pneumatic and electro-pneumatic circuits. Introduction to PLC –PLC programming for automation applications.

4. Automobile Engineering lab –9 hours

Overview of automobiles – components –functioning of various sub-systems; Power train, steering system, suspension system and braking system. Introduction to electric vehicles, hybrid vehicles, alternate fuels. Introduction to E Mobility.

Unit II

Introduction to Raspberry Pi, GPIO programming, interfacing sensors and actuators. Introduction to IoT, communicating sensor data to cloud platforms.

Glimpses of Glorious India

Syllabus

Syllabus

Face the Brutes, Role of Women in India, Acharya Chanakya, God and Iswara, Bhagavad Gita: From Soldier to Samsarin to Sadhaka, Lessons of Yoga from Bhagavad Gita, Indian Soft powers, Preserving Nature through Faith, Ancient Indian Cultures (Class Activity), Practical Vedanta, To the World from India, Indian Approach to Science.

CSE-B TIME TABLE												
Master Time Table CSE II Semester B Section												
Dept. CSE				Class Room Venue:511								
Class: B. Tech. CSE-B Semester-II												
In-charge Advisors: Name: Dr. Karmala Govri Raghavendra Narayan Email : kgr_narayan@av.amrita.edu Mobile Number : +91-9705409393				2. Name: Dr. Ankit Sachan Email : a_ankit@av.amrita.edu Mobile Number : +91-8381800129								
TIME/DAY	Slot 1 08:55-09:45am		Slot 2 09:45-10:35am	10:35am- 10:45am	Slot 3 10:45-11:35am	Slot 4 11:35am-12:25pm	Slot 5 12:25am-01:15pm	01:15pm- 02:05pm	Slot 6 02:10-03:00pm	3:00pm- 3:10pm	Slot 7 03:10-04:00pm	Slot 8 04:00-04:50pm
Monday	23MAT117 Linear Algebra LAB(BYOD)			Short Break	22ADM111 Glimpses of Glorious India	23PHY115 Modern Physics	LIBRARY	Lunch Break	23MEE115 Manufacturing Practice LAB	Short Break	23MEE115 Manufacturing Practice LAB	
Tuesday	23MAT116 Discrete Mathematics	23MAT117 Linear Algebra	23PHY115 Modern Physics		23CSE111 Object Oriented Programming	23CSE113 User Interface Design	22ADM111 Glimpses of Glorious India(P)		Library/Sport			
Wednesday	23CSE111 Object Oriented Programming_lab2				23MAT116 Discrete Mathematics	23CSE113 User Interface Design	22ADM111 Glimpses of Glorious India		Library/Sports			
Thursday	23MAT117 Linear Algebra	23CSE111 Object Oriented Programming	23MAT116 Discrete Mathematics		COUNSELLING	LIBRARY	23PHY115 Modern Physics(T)		Library/Sport			
Friday	23CSE113 User Interface Design Lab BYOD				23MAT117 Linear Algebra	23CSE111 Object Oriented Programming	Library/Sports		COUNSELLING		23MAT116 Discrete Mathematics_lab_lab1	
Course Code	Course Name				L-T-P	credits	Faculty Name				Assisting Faculty	
23MAT116	Discrete Mathematics				3-0-2	4	Dr.Srivivasa Rao Thota					
23CSE111	Object Oriented Programming				3-0-2	4	Dr. Balaji TK					
23PHY115	Modern Physics				2-1-0	3	Dr. C R Kesavulu					
23CSE113	User Interface Design				2-0-2	3	Dr.Rajkumar Batchu				Mr. Barge Bhandwaj	
23MEE115	Manufacturing Practice				0-0-3	1	Dr. Sathies S.					
22ADM111	Glimpses of Glorious India				2-0-1	2	Dr. Siva Panuganti					
	Total (15L + 1Tut +12P =28 hrs)				15-1-12	21						

Explanation:

This program is an HTML webpage that shows the 2nd semester syllabus and CSE-B timetable in one page. It starts with the basic HTML structure and sets the page title. CSS is used to add borders and collapse lines for all tables so they look neat. The page background color is set to orange, and a centered heading “SYLLABUS” is displayed at the top.

Next, the program lists each subject name using <h2> headings and shows the syllabus images using tags. The images are loaded from a folder named SY_IMG and displayed with a fixed width, so each subject’s syllabus appears visually on the page.

After the syllabus section, the program creates a large timetable using an HTML table. It uses rows and columns to show class details, advisors, time slots, subjects, labs, breaks, and faculty information. Features like colspan and rowspan are used to merge cells for breaks and lab hours, and some cells are colored to highlight breaks and counselling. Overall, the program combines images and tables to present both syllabus content and the weekly timetable in a structured format.

Lab-4

4a-

Aim: A college wants to collect student details through an online admission form. Design a registration form using only HTML tags(no css) with proper labels, table alignment ,background, colour and borders.

Procedure:

Code:

```
<!DOCTYPE html>
<html>
  <head>
    <style>
      table, th, td {
        border: 1px solid black;
        background-color: lightyellow;
      }
    </style>
    <Title>registration form</form></Title>
  </head>
  <body style="background-image:url(bg.2.png);background-size: auto;">
    <forms>
      <center>
        <table style="width: 50%" style="text-align: center;">
          <h1>registration form</h1>
          <tr>
            <td><label id="fname">FullName</label></td>
            <td><input type="name" for="fname" placeholder="name"></td>
          </tr>
          <tr>
            <td><label id="email">Email</label></td>
            <td><input type="email" for="email" placeholder="email"></td>
          </tr>
          <tr>
            <td><label id="pnumber">PhoneNumber</label></td>
            <td><input type="number" for="pnumber" placeholder="Number"></td>
          </tr>
          <tr>
            <td><label type="radio" id="Gender">Gender</label></td>
```

```

        <td><input type="radio" name="Gender" for="Gender">Male
        <input type="radio" name="Gender" for="Gender">Female
        <input type="radio" name="Gender" for="Gender">Others
        </td>
    </tr>
    <tr>
        <td><label for="Course">Course Applying For</label></td>
        <td>
            <select name="Course" id="Course">
                <OPTION value="B-tech">B-tech</OPTION>
                <option value="B-arch">B-arch</option>
                <option value="Degree">Degree</option>
                <option value="Agriculture">Agriculture</option>

            </select>
        </td>
    </tr>
    <tr>
        <td><label type="radio" id="Hobbies">Hobbies</label></td>
        <td><input type="radio" id="Hobbies" for="Hobbies">Music
        <input type="radio" id="Sports" for="Hobbies">Sports
        <input type="radio" id="Reading" for="Hobbies">Reading
        </td>
    </tr>
    <tr>
        <td><label for="dob">Date Of Birth</label></td>
        <td>
            <input type="date" id="dob" name="dob">
        </td>
    </tr>
    <tr>
        <td><label for="file" id="file">Uploding Certificates</label></td>
        <td><input type="file" for="file"></td>
    </tr>
    <tr>
        <td><label for="address">Address</label></td>
        <td><textarea id="address" rows="11" name="address"></textarea></td>

    </tr>
    <tr aria-colspan="2">
        <td style="text-align: center;" colspan="2">
            <input type="submit" value="Submit Form">

```

```

        <input type="reset" value="Reset Form">
    </td>
</tr>

</table>
</center>

</forms>

</body>
</html>

```

Output:

The screenshot shows a web browser window with a file path in the address bar. The page displays a registration form with the following fields and controls:

- Full Name:** Text input field with placeholder 'name'.
- Email:** Text input field with placeholder 'email'.
- PhoneNumber:** Text input field with placeholder 'Number'.
- Gender:** Radio buttons for Male, Female, and Others.
- Course Applying For:** Dropdown menu with 'B-tech' selected.
- Hobbies:** Radio buttons for Music, Sports, and Reading.
- Date Of Birth:** Date picker showing 'dd-mm-yyyy'.
- Uploading Certificates:** File upload control showing 'Choose File' and 'No file chosen'.
- Address:** Large text area for multi-line input.
- Buttons:** 'Submit Form' and 'Reset Form' buttons at the bottom.

Explanation:

This HTML program creates a registration form using some important tags and attributes. The <head> section includes the <style> for tables, td, th, and background colours are defined. In <body> the <table> tag is used to structure the form, with <tr> for rows and <td> for cells. Labels are created with <label> and connected to inputs using the for attribute, while user inputs are taken through <input> with types such as text/name, email, number, radio, date, file, submit, and reset. Dropdown selection is created using <select> and <option> tags, and multi-line text input is handled by <textarea> with rows and name attributes. Table formatting uses attributes like border, width, and colspan, while common input attributes include id, name, placeholder, value, and type, which control identification, grouping, hint text, displayed button text, and input is used.

4b-

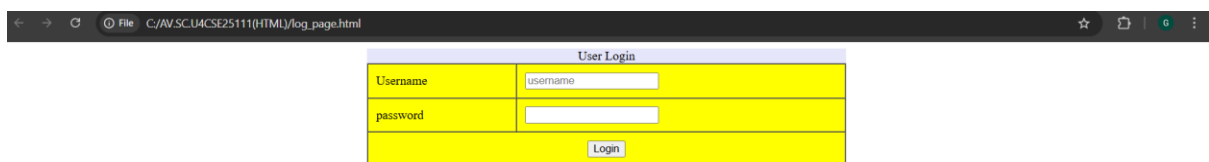
Aim: A company wants to design a web page that displays two different forms side by side using “iframes”. One iframe should display a login form and the other should display a register form.

Program:

Code-1: User Login

```
<!DOCTYPE html>
<html>
  <head>
    <title>Login Page</title>
  </head>
  <body>
    <form >
      <table style="background-color: yellow;" align="center" border="1"
cellpadding="10px" cellspacing="0" width="40%" >
        <caption style="background-color: lavender;">User Login </caption>
        <tr>
          <td><label for="Username">Username</label></td>
          <td><input type="text" placeholder="username" id="username"></td>
        </tr>
        <tr>
          <td><label for="password">password</label></td>
          <td><input type="password" id="password"></td>
        </tr>
        <tr>
          <td colspan="2" align="center"><button>Login</button></td>
        </tr>
      </table>
    </form>
  </body>
</html>
```

Output:

A screenshot of a web browser window showing a "User Login" form. The browser's address bar displays the file path "C:/AV/SC/U4CSE25111(HTML)/log_page.html". The form is titled "User Login" and is styled with a yellow background and a lavender caption. It contains two input fields: "Username" and "password", both with placeholder text. Below the input fields is a "Login" button. The form is centered on the page.

User Login	
Username	username
password	
Login	

Explanation:

This HTML program creates a simple User login page using basic structure and form elements. The <body> tag holds all visible content. A <form> tag is used to collect user input. In <table> newly we use the cellpadding to give inner spacing, cellspacing to control cell gaps. The <caption> tag provides a table title (“User Login”). Each row is created with <tr>, and cells are created with <td>. The <label> tag describes each input field and connects using the for attribute, while <input> fields use the type attribute such as text for username and password for 42iding password entry, along with attributes like id and placeholder. The login action button is created using the <button> tag, and colspan is used to merge two columns.

Code-2: User Registration

```
<!DOCTYPE html>
<html>
  <head>
    <title> User registration</title>
  </head>
  <body>
    <center>
      <forms>
        <table border="black " cellspacing="3px" cellpadding="5px" style="background-
color: yellow;">
          <caption style="background-color: lavender;">User registration</caption>
          <tr>
            <td>
              <label for="Newuser">Username:</label>
            </td>
            <td>
              <input type="text" id="Newuser" name="Newuser">
            </td>
          </tr>
          <tr>
            <td>
              <label for="Email">Email:</label>
            </td>
            <td>
              <input type="email" id="Email" name="Email">
            </td>
          </tr>
          <tr>
            <td>
              <lable id="pnumber">Phone Number:</lable>
```

```

        </td>
        <td>
            <input type="number" for="pnumber">
        </td>
    </tr>
    <tr>
        <td>
            <label for="Password">Password:</label>
        </td>
        <td>
            <input type="password" id="newpassword" name="Password">
        </td>
    </tr>
    <tr>
        <td style="text-align: center;" colspan="2">
            <input type="Submit" value="Register">

            <input type="reset" value="reset">
        </td>
    </tr>
</table>
</forms>
</center>
</body>
</html>

```

Output:

The screenshot shows a web browser window with the address bar displaying 'C:/AV/SCU4CSE25111(HTML)/submit_form.html'. The browser content shows a form titled 'User registration'. The form contains four labeled input fields: 'Username:', 'Email:', 'Phone Number:', and 'Password:'. Below these fields are two buttons labeled 'Register' and 'reset'.

Explanation:

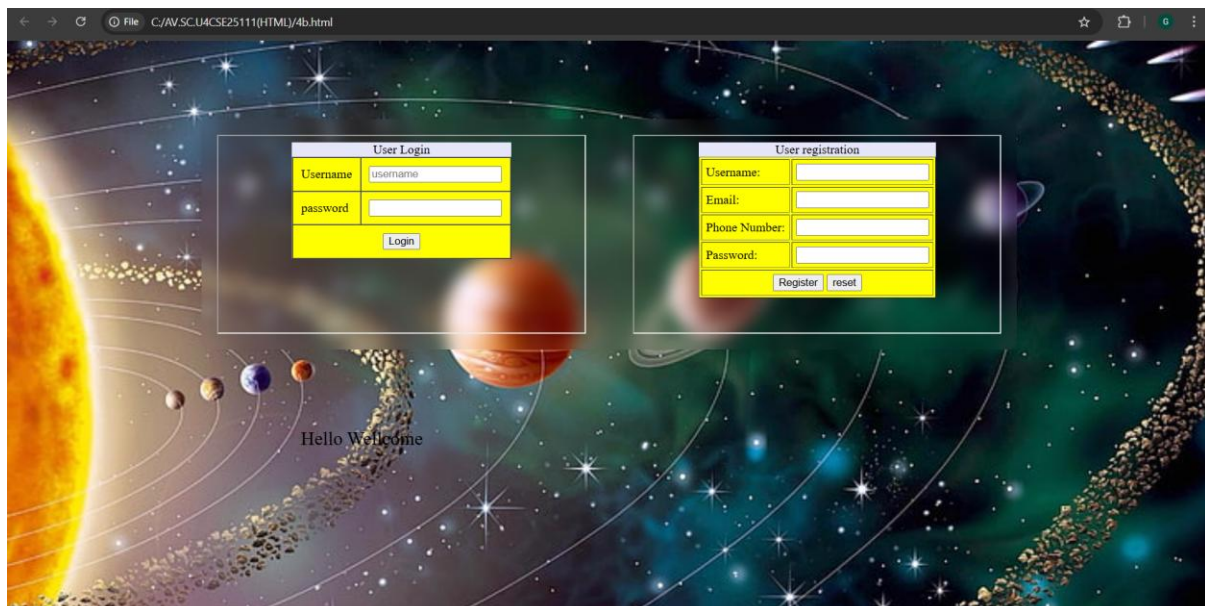
This HTML program creates a user registration page using basic structure and form elements. The <caption> tag gives the table a heading. Each row is defined with <tr> and cells with <td>. Field names are shown using the <label> tag with the for attribute, and user inputs are collected using <input> elements with types such as text, email, number, and password, along with attributes like id and name for identification. The submit and reset actions are created using <input type="submit"> and <input type="reset">. So these are used in these code.

Code-3:Using “iframe” two forms display on web page.

```
<!DOCTYPE html>
<html>
  <head>
    <title>loginpage</title>
  </head>
  <body style="background-image: url(space_image.png);background-repeat: no-repeat;background-size: cover;">

    <div style="display:flex; gap:60px; min-height: 80px;margin: 100px auto; padding:20px; background-color: rgba(255, 255, 255, 0.2);backdrop-filter: blur(10px);border-radius: 10px;color: black ;max-width: 1000px;">
      <iframe src=file:///C:/AV.SC.U4CSE25111(HTML)/log_page.html width="3px" height="250px" style="flex: 1; text-align: center;"></iframe>
      <iframe src=file:///C:/AV.SC.U4CSE25111(HTML)/submit_form.html width="4px" height="250px" style="flex: 1;text-align: center;"></iframe>
    </div>
    <marquee direction="right" style="font-size: x-large;">Hello Wellcome</marquee>
  </body>
</html>
```

Output:



Explanation:

The code is done by above two codes. Combining the two codes in same web page using the `<iframe>` tag. This HTML program creates a styled login container page that displays two embedded pages side-by-side. A `<div>` tag is used as a layout container to group elements and arrange them using CSS Flexbox with properties such as `display:flex` for horizontal layout. Two `<iframe>` tags are used to embed other HTML pages inside the container, where the `src` attribute specifies the file path and width, height, and style control frame size and flex behavior. Finally, the `<marquee>` tag creates scrolling text using the `direction` attribute and font styling, though it is deprecated in modern HTML.

LAB-5

5(A)

AIM:

To create a responsive card layout using HTML and CSS with three cards, each containing an image, title, and button.

PROGRAM:

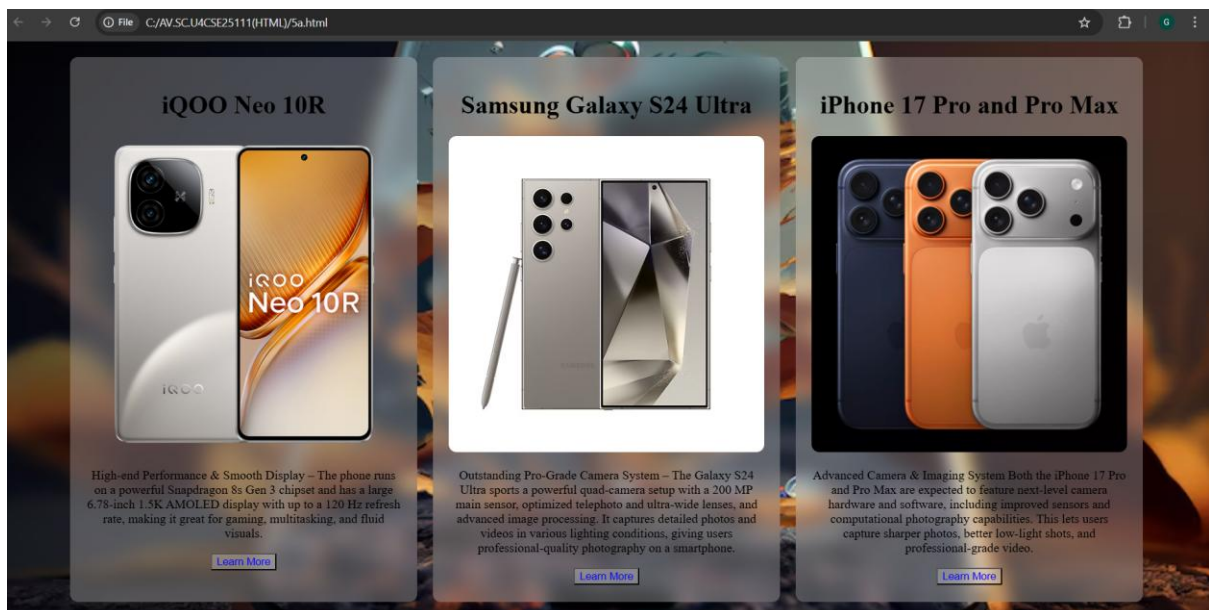
```
body{
  background-color: lightcoral;
  background-image: url('https://webfiles.amrita.edu/2020/05/IlZJZE8c-amrita-vishwa-vidyapeetham-coimbatore-campus-dron-image-01-2.jpg');
  background-repeat: no-repeat;
  background-size: cover;
  background-position: center;
  background-attachment: scroll;
}
.button_container{
  display: flex;
  justify-content: center;
  margin-top: 20px;
  background-color: rgba(255, 255, 255, 0.25);
  height: 50px;
  border-radius: 10px;
  backdrop-filter: blur(5px);
}
.button{
  margin: 10px 10px 10px 10px;
  width: 50%;
  padding-right: 10px;
}
a{
  text-decoration: none;
  color: red;
  font-size: x-large;
}
a:hover{
  color: yellow;
  font-size: x-large;
}
iframe{
  width: 100%;
  height: 600px;
  border: none;
}
header{
  display: flex;
```

```

justify-content: center;
width: 100%;
img{
  width: 20%;
}
}
.frame_container{
  display: flex;
  justify-content: center;
  align-items: center;
}
}
.var{
  width: 500px;
  min-height: 400px;
  margin-left: 100px;
  margin-top: 10px;
  background: rgba(255,255,255,0.92);
  padding: 30px;
  max-width: 500px;
  border-radius: 15px;
  box-shadow: 0 8px 25px rgba(0,0,0,0.2);
  line-height: 1.6;
  font-size: 17px;
  color: #333;
  overflow: auto;
}
}
.const{
  margin-right: 100px;
}
}

```

OUTPUT:



EXPLANATION:

This HTML program creates a responsive card layout to showcase smartphones including iQOO Neo 10R, Samsung Galaxy S24 Ultra, and iPhone 17 Pro. The `<!DOCTYPE html>` declaration defines the HTML5 document type, and the `<html>` tag is the root element. Inside the `<head>`, the `<title>` tag sets the browser tab name, and the `<style>` tag contains internal CSS for page design. The `<body>` uses CSS properties like `background-image`, `background-repeat`, `background-size`, `background-position`, `margin`, and `height` to control the full-page background. A `<div>` with class `card_container` uses Flexbox properties such as `display: flex`, `flex-wrap`, `justify-content`, `align-items`, and `align-content` to create a responsive layout. Each `.card` div acts as a content box with styling like `width`, `height`, `background-color` using `rgba` transparency, `backdrop-filter` for blur effect, `border-radius`, `padding`, `margin`, `overflow`, and `text-align`. The `<img>` tag displays product images using properties like `width`, `height`, `object-fit`, and `border-radius`. Text content is structured using `<h1>` for headings and `<p>` for descriptions. The `<a>` (anchor) tag with `href` and `target="_blank"` is used to link external product pages, and the `<button>` tag creates interactive buttons with hover effects using the `:hover` pseudo-class.

5(B)

Aim:

To design a mini website using iframe with navigation links (Home, About, Contact), where clicking on each link displays the respective content within the same page using internal or external CSS.

PROGRAM:

CSS-CODE:

```
h1 {
  color:red;
  text-align: center;
  font-family: 'Lucida Sans', 'Lucida Sans Regular', 'Lucida Grande', 'Lucida Sans Unicode',
  Geneva, Verdana, sans-serif;
  font-style:italic
}
pre{
  color: yellow;
  text-align: center;
  font-family: 'Lucida Sans', 'Lucida Sans Regular', 'Lucida Grande', 'Lucida Sans Unicode',
  Geneva, Verdana, sans-serif;
  font-size: 25px;
}
h2{
  color:red;
  text-align: center;
}
p{
  color:yellow;
  text-align: center;
  font-family: 'Lucida Sans', 'Lucida Sans Regular', 'Lucida Grande', 'Lucida Sans Unicode',
  Geneva, Verdana, sans-serif;
  font-size: 30px;
}
```

CODE:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Index page</title>
    <style>
      body {
        background-image: url('5b.png');
        background-size: cover;
        background-color: black;
```

```

background-position: center;
background-repeat: no-repeat;
margin: 0;
height: 100vh;
border: none;
}
.frame{
  align-self: center;
  width: 100%;
  height: 600px;
  margin-right: 20px ;
  border: none;
  outline: none;
}
.button_container{
  display: flex;
  justify-content: center;
  padding: 10px;
  margin: 10px 10px 0px 10px;
}
.button{
  color: #E0E0E0;
  margin: 10px 10px 0px 10px;
}
.frame_container{
  display: flex;
  justify-content: center;
  align-items: center;
  min-height: 100vh;
}
</style>
</head>
<body>
  <div class="button_container">
    <div class="button">
      <a href="Home.html" target="content">
        <button>Home</button>
      </a>
    </div>
    <div class="button">
      <a href="About.html" target="content">
        <button>About</button>
      </a>
    </div>
    <div class="button">
      <a href="Contacts.html" target="content">
        <button>Contact</button>
      </a>
    </div>
  </div>

```

```

    </div>
</div>
<div class="frame_container">
  <iframe name="content" class="frame" src="Home.html"></iframe>
</div>
</body>
</html>

```

OUTPUT:

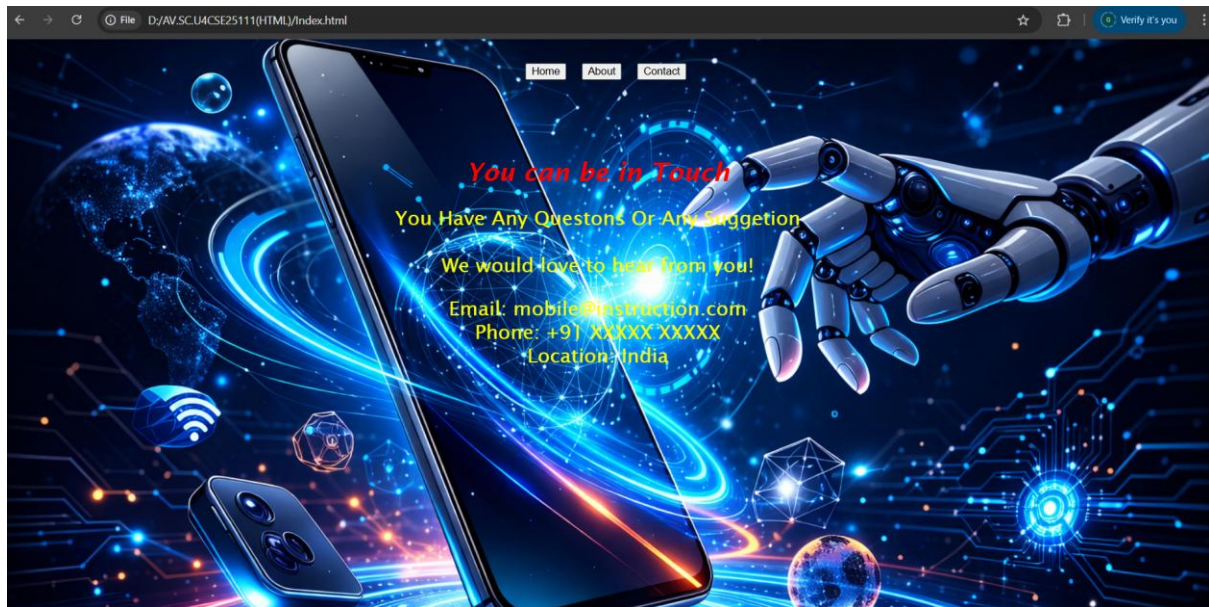
HOMEPAGE -



ABOUTPAGE -



CONTACTPAGE –



EXPLANATION:

This HTML program creates an index page with navigation buttons and an embedded content area using frames. The `<!DOCTYPE html>` declaration defines the HTML5 document type, and the `<html>` tag is the root element of the webpage. Inside the `<head>` section, the `<title>` tag sets the browser tab name as "Index page," and the `<style>` tag contains internal CSS used to design the layout. The `<body>` tag holds all visible content and applies background styling using CSS properties such as `background-image`, `background-size`, `background-position`, `background-repeat`, `margin`, and `height`. The `<div>` elements with classes like `button_container`, `button`, and `frame_container` are used to group and organize page elements. CSS Flexbox properties such as `display: flex`, `justify-content`, `align-items`, `padding`, and `margin` help align buttons and center the frame area. Navigation is created using the `<a>` (anchor) tag with the `href` attribute to link different pages like Home, About, and Contact, and the `target` attribute directs the page to load inside a specific frame. The `<button>` tag is used to create clickable navigation buttons. Finally, the `<iframe>` tag embeds another HTML page inside the index page using attributes like `name`, `class`, and `src`, allowing the selected page to load dynamically within the frame area.

5(C)

Aim:

To design an Amrita login page using HTML and CSS that displays different campuses, where clicking on each campus shows its details on the same page.

PROGRAM:

CSS-CODE:

```
body{
  background-color: lightcoral;
  background-image: url('https://webfiles.amrita.edu/2020/05/IlZJZE8c-amrita-vishwa-vidyapeetham-coimbatore-campus-dron-image-01-2.jpg');
  background-repeat: no-repeat;
  background-size: cover;
  background-position: center;
  background-attachment: scroll;
}
.button_container{
  display: flex;
  justify-content: center;
  margin-top: 20px;
  background-color: rgba(255, 255, 255, 0.25);
  height: 50px;
  border-radius: 10px;
  backdrop-filter: blur(5px);
}
.button{
  margin: 10px 10px 10px 10px;
  width: 50%;
  padding-right: 10px;
}
a{
  text-decoration: none;
  color: red;
  font-size: x-large;
}
a:hover{
  color: yellow;
  font-size: x-large;
}
iframe{
  width: 100%;
  height: 600px;
  border: none;
}
header{
  display: flex;
```

```

        justify-content: center;
        width: 100%;
        img{
            width: 20%;
        }
    }
    .frame_container{
        display: flex;
        justify-content: center;
        align-items: center;
    }
    .var{
        width: 500px;
        min-height: 400px;
        margin-left: 100px;
        margin-top: 10px;
        background: rgba(255,255,255,0.92);
        padding: 30px;
        max-width: 500px;
        border-radius: 15px;
        box-shadow: 0 8px 25px rgba(0,0,0,0.2);
        line-height: 1.6;
        font-size: 17px;
        color: #333;
        overflow: auto;
    }
    .const{
        margin-right: 100px;
    }

```

CODE:

```

<!DOCTYPE html>
<html>
<head>
    <title>Amrita</title>
    <link rel="stylesheet" href="main_style.css">
</head>
<body>
    <header>
        
    </header>
    <div class="button_container">
        <div class="button">
            <a href="home.html" target="content">home</a>
        </div>
    </div>

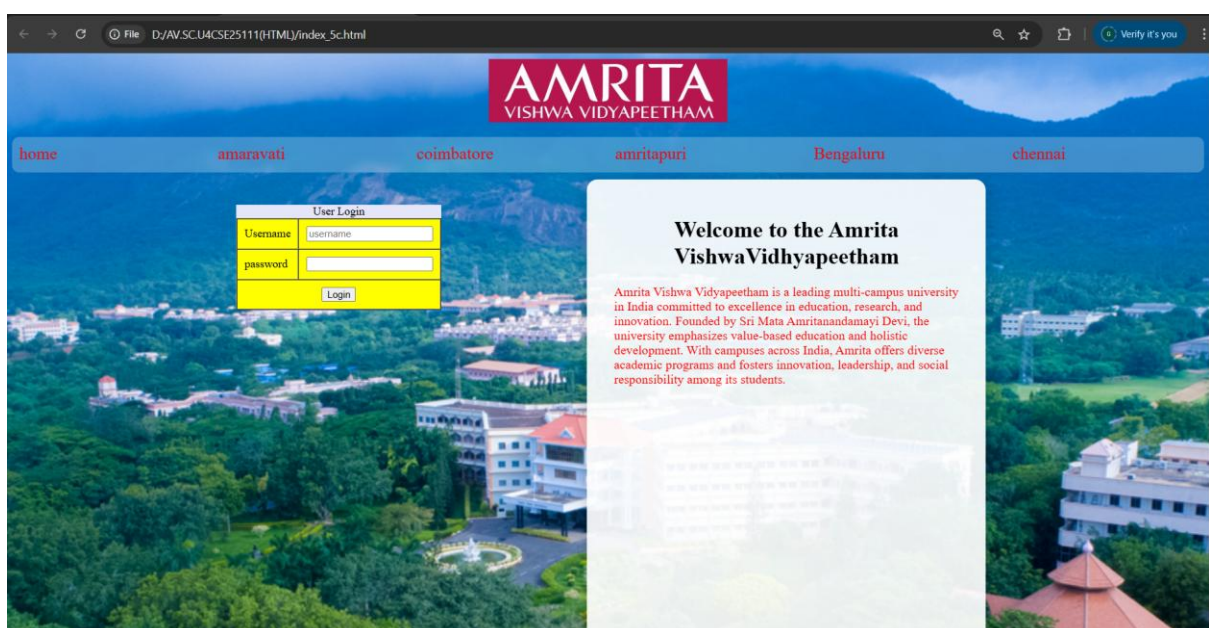
```

```

<div class="button">
  <a href="amaravati.html" target="content">amaravati</a>
</div>
<div class="button">
  <a href="coimbatore.html" target="content">coimbatore</a>
</div>
<div class="button">
  <a href="amritapuri.html" target="content">amritapuri</a>
</div>
<div class="button">
  <a href="bengaluru.html" target="content">Bengaluru</a>
</div>
<div class="button">
  <a href="chennai.html" target="content">chennai</a>
</div>
</div>
<div class="frame_container">
  <div class="const">
    <iframe src="log_page.html" class="frame"></iframe>
  </div>
  <div class="var">
    <iframe class="frame" name="content" src="home_5c.html"></iframe>
  </div>
</div>
</body>
</html>

```

OUTPUT:



EXPLANATION:

This HTML program creates a structured webpage for Amrita Vishwa Vidyapeetham with navigation and embedded content areas. The `<!DOCTYPE html>` declaration defines the HTML5 document type and the `<html>` tag is the root element of the webpage. Inside the `<head>` section, the `<title>` tag sets the browser tab name as “Amrita,” and the `<link>` tag with attributes `rel="stylesheet"` and `href` connects an external CSS file (`main_style.css`) for styling the page. The `<body>` contains the visible content, starting with the `<header>` tag that displays a university logo using the `<img>` tag with attributes like `src` for the image source and `alt` for alternative text. Several `<div>` elements such as `button_container` and `button` are used to organize navigation links. Navigation is created using the `<a>` (anchor) tag with the `href` attribute linking to different campus pages like Amaravati, Coimbatore, Amritapuri, Bengaluru, and Chennai, while the `target` attribute loads these pages inside a specific frame. The layout also includes containers like `frame_container`, `const`, and `var` to manage page structure. Two `<iframe>` elements are used to embed other HTML pages inside the main page: one loads a login page and the other loads dynamic content pages using attributes like `src`, `class`, and `name`, allowing users to switch between campus pages without leaving the main layout.

LAB-6

6A.

AIM:

Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books. Based on the category we select corresponding products should be displayed on the sidebar.

PROGRAM:

CSS CODE:

```
*{
margin:0;
padding:0;
box-sizing:border-box;
font-family:'Times New Roman', Times, serif;
}

body{
background:#000;
color:white;
overflow-x:hidden;
}

/* animated falling lines background */

/* .background{
position:fixed;
width:100%;
height:100%;
background-image:linear-gradient(transparent 95%,rgba(255,255,255,0.2) 100%);
background-size:3px 40px;
animation:rain 1s linear infinite;
z-index:-1;
}

@keyframes rain{
0%{background-position:0 0;}
100%{background-position:0 40px;}
} */

.background{
position: fixed;
width: 100%;
height: 100%;
```

```

    background: linear-gradient(rgba(0,0,0,0.6),
    rgba(0,0,0,0.6)),url("https://www.shutterstock.com/image-photo/many-various-modern-
gadgets-on-260nw-2661177029.jpg");
    background-size: cover;
    background-position: center;
    background-repeat: no-repeat;
    z-index: -1;
}

/* MAIN PAGE GRID */

.container{
display:grid;
grid-template-columns:240px 1fr;
grid-template-rows:auto 1fr auto;
grid-template-areas:
"header header"
"sidebar courses"
"footer footer";
min-height:100vh;
}

/* HEADER */

.header{
grid-area:header;
background:#111;
padding:20px;
text-align:center;
border-bottom:1px solid #333;
}

.sale{
color:#00ff88;
font-size:14px;
margin-top:5px;
}

/* SIDEBAR */

.sidebar{
grid-area:sidebar;
background:#111;
padding:20px;
border-right:1px solid #333;
}

.sidebar h3{

```

```

margin-bottom: 15px;
}

.sidebar ul{
list-style: none;
}

.sidebar li{
margin: 12px 0;
color: #ccc;
cursor: pointer;
}

.sidebar li: hover{
color: #00ffcc;
}

/* COURSES GRID */

.courses{
grid-area: courses;
display: grid;
grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
gap: 25px;
padding: 30px;
}

/* COURSE CARD */

.card{
background: rgba(20,20,20,0.9);
border-radius: 12px;
overflow: hidden;
text-align: center;
padding-bottom: 15px;
transition: 0.3s;
border: 1px solid rgba(255,255,255,0.1);
}

.card: hover{
transform: translateY(-8px) scale(1.03);
box-shadow: 0 0 15px rgba(0,255,200,0.4);
}

.card img{
width: 100%;
height: 140px;
object-fit: cover;
}

```

```

}

.card h4{
margin-top:10px;
}

.price{
margin:10px 0;
font-weight:bold;
}

.price span{
text-decoration:line-through;
color:red;
margin-left:8px;
}

/* FOOTER */

.footer{
grid-area:footer;
background:#111;
display:grid;
grid-template-columns:repeat(3,1fr);
gap:20px;
padding:25px;
border-top:1px solid #333;
}

.footer h4{
margin-bottom:10px;
}

```

MAIN CODE:

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Byte Bazaar</title>
<link rel="stylesheet" href="6a.css">
</head>

<body>

<div class="background"></div>

<div class="container">

```

```
<header class="header">
<h1>Byte Bazaar</h1>
<div class="sale">FLASH SALE: UP TO 40% OFF ON ALL ELECTRONIC
GADGETS</div>
</header>

<aside class="sidebar">
<h3>Categories</h3>
<ul>
<li>All</li>
<li>Mobile Phones</li>
<li>Laptops</li>
<li>Smartwatches</li>
<li>Headphones</li>

</ul>
</aside>

<main class="courses">

<div class="card">

<h4>Iphone 16 pro</h4>
<p class="price">₹1,00,490 <span>₹1,11,999</span></p>
<p>Deal ends in: 2 days</p>
</div>

<div class="card">

<h4>Lenovo Legion Pro 5 2025</h4>
<p class="price">₹2,00,827 <span>₹2,65,890</span></p>
<p>Deal ends in: 12 hours</p>
</div>

<div class="card">

<h4>Samsung Galaxy Tab S11 Ultra</h4>
<p class="price">₹1,12,998 <span>₹1,33,999</span></p>
<p>Deal ends in: 19 hours</p>
</div>

<div class="card">

<h4>Apple Watch Ultra 2</h4>
<p class="price">₹91,799 <span>₹1,32,651</span></p>
<p>Deal ends in: 1 days</p>
</div>

<div class="card">

<h4>JBL Tune 770NC Wireless</h4>
<p class="price">₹4,999 <span>₹9,999</span></p>
<p>Deal ends in: 3 days</p>
</div>

<div class="card">

```

```


<h4>Dell Pro Premium Active Pen</h4>
<p class="price">₹4,899 <span>₹8,499</span></p>
<p>Deal ends in: 3 days</p>
</div>

</main>

<footer class="footer">

<div>
<h4>Amrita Vishwa Vidyapeetam</h4>
<p>support@avamrita.com</p>
</div>

<div>
<h4>Developer</h4>
<p>Gowrneni Mokshagnasai</p>
<p>ID: CSE25111</p>
<p>Section: CSE-B</p>
</div>

<div>
<h4>Quick Links</h4>
<p>GitHub</p>
<p>Instagram</p>
<p>Amrita Vishwa Vidyapeetam, Amaravati</p>
</div>

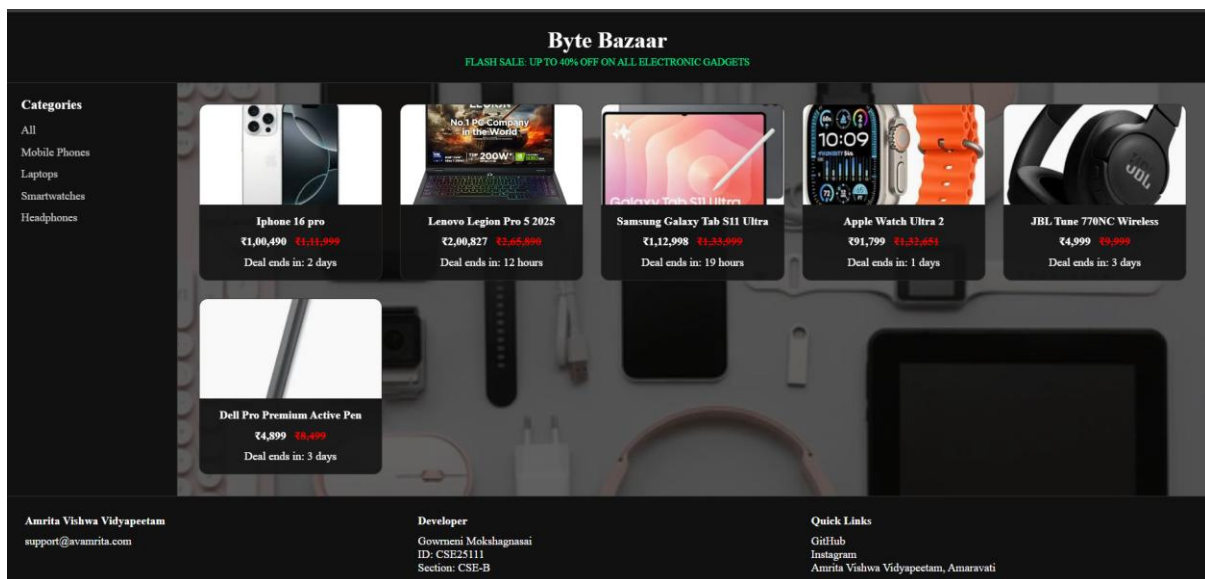
</footer>

</div>

</body>
</html>

```


OUTPUT:



EXPLANATION:

This HTML program creates an online electronics shopping webpage called Byte Bazaar that displays different gadgets in a card layout. The `<!DOCTYPE html>` declaration defines the HTML5 document type and the `<html>` tag is the root element of the webpage. Inside the `<head>` section, the `<meta charset="UTF-8">` tag defines the character encoding, the `<title>` tag sets the browser tab name, and the `<link>` tag with attributes `rel="stylesheet"` and `href` connects an external CSS file (6a.css) for styling. The `<body>` contains the main webpage structure including a background container `<div>` and a main `<div class="container">` used to group all elements. The `<header>` section displays the website title and a promotional banner using `<h1>` and `<div>` tags. A `<aside>` tag creates a sidebar that lists product categories using `<ul>` and `<li>` elements. The main product area uses the `<main>` tag with multiple `<div class="card">` elements to display products such as iPhone 16 Pro, Lenovo Legion Pro 5, Samsung Galaxy Tab S11 Ultra, Apple Watch Ultra 2, and JBL Tune 770NC. Each card contains an `<img>` tag to display the product image, `<h4>` for the product name, and `<p>` elements to show the price and deal time, with `<span>` used to display the original price beside the discounted price. Finally, the `<footer>` section displays additional information such as university name, developer details, and quick links using `<div>`, `<h4>`, and `<p>` tags, completing the webpage layout.

6B.

AIM : Using switches.

CODE:

```
<!DOCTYPE html>
<html>
  <head>
    <title>E-Commerce store</title>
    <style>
      body{
        font-family: Arial, sans-serif;
        margin: 0;
        padding: 0;
        background-color: grey;
      }
      .container{
        display: grid;
        grid-template-columns:200px 1fr;
        grid-template-rows:auto auto 1fr auto;
        gap:20px;
        padding:20px;
        min-height:100vh;
        grid-template-areas:
          "header header"
          "sale sale"
          "sidebar products"
          "footer footer";
      }
      .header{
        grid-area: header;
        background-color: red;
        color: white;
        padding: 10px;
        text-align: center;
      }
      .sale{
        grid-area: sale;
        background-color:black;
        color: white;
        padding: 20px;
        text-align: center;
      }
      .sidebar{
        grid-area: sidebar;
        background-color: lightgray;
        padding: 20px;
```

```

}
.sidebar ul{
  list-style:none;padding:0;
}
.sidebar li{
  margin:15px 0;
}
.sidebar a{
  color:#fff;
  text-decoration:none;
  font-size:18px;
  transition:.3s;
}
.sidebar a:hover{
  color:#00ffd5;padding-left:8px;
}
.main{
  grid-area:products;
}
iframe{
  width:100%;
  height:100%;
  border:0;
}
.products{
  grid-area: products;
  background-color: white;
  padding: 0;
}
.card{
  background-color:red;
  color: white;
  padding: 20px;
  border-radius: 5px;
  text-align:center;
}
.button{
  display: inline-block;
  padding: 10px 20px;
  background-color: palevioletred;
  color: white;
  text-decoration: none;
  border-radius: 5px;
  margin-top: 10px;
}
.button:hover{
  background-color: plum;
}

```

```

.footer{
  grid-area: footer;
  background-color: lightgray;
  color: black;
  padding: 20px;
  text-align: center;
}

</style>
</head>
<body>
  <div class="container">

    <div class="header">
      <h1>Welcome To GMS Store</h1>
    </div>

    <div class="sale">
      <h2>Big Sale! Up to 50% Off!</h2>
    </div>

    <div class="sidebar">
      <h3>Categories</h3>
      <ul>
        <li><a href="electronics.html" target="contentFrame">Electronics</a></li>
        <li><a href="clothings.html" target="contentFrame">Clothing</a></li>
        <li><a href="home-kitchen.html" target="contentFrame">Home &
Kitchen</a></li>
        <li><a href="books.html" target="contentFrame">Books</a></li>
      </ul>
    </div>

    <div class="products">
      <iframe name="contentFrame"></iframe>
    </div>

    <div class="footer">
      <p>© 2026 GMSAI Store</p>
    </div>

  </div>
</body>
</html>

```

HOME KITCHEN CODE:

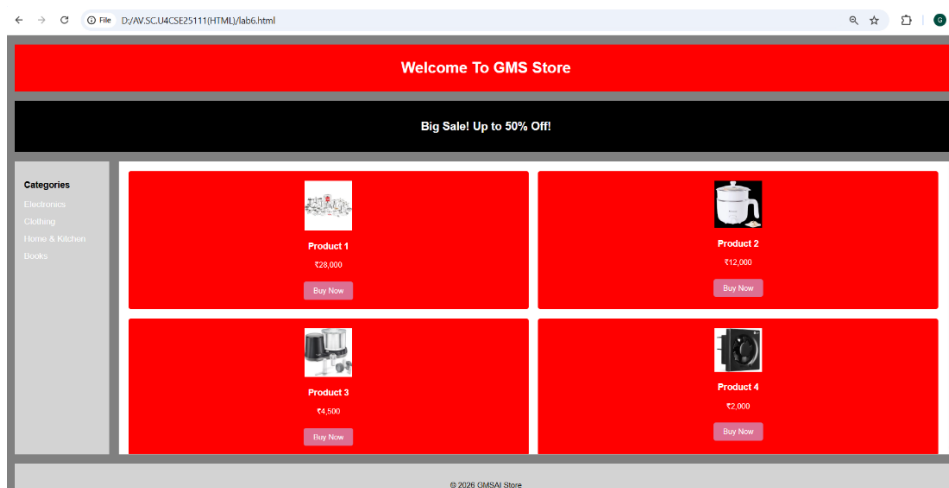
```
<!DOCTYPE html>
<html>
  <head>
    <style>
      body{
        font-family: Arial, sans-serif;
        margin: 0;
        padding: 0;
        background-color: grey;
      }
      .products{
        grid-area: products;
        background-color: white;
        padding: 20px;
        display: grid;
        grid-template-columns: repeat(2, 1fr);
        gap: 20px;
      }
      .card{
        background-color:red;
        color: white;
        padding: 20px;
        border-radius: 5px;
        text-align:center;
      }
      .button{
        display: inline-block;
        padding: 10px 20px;
        background-color: palevioletred;
        color: white;
        text-decoration: none;
        border-radius: 5px;
        margin-top: 10px;
      }
      .button:hover{
        background-color: plum;
      }
    </style>
  <body>
    <div>
      <div class="products">
        <div class="card">
          
          <h3>Product 1</h3>
        </div>
      </div>
    </div>
  </body>
</html>
```

```

        <p>₹28,000</p>
        <a href="#" class="button">Buy Now</a>
    </div>
    <div class="card">
        
        <h3>Product 2</h3>
        <p> ₹12,000</p>
        <a href="#" class="button">Buy Now</a>
    </div>
    <div class="card">
        
        <h3>Product 3</h3>
        <p> ₹4,500</p>
        <a href="#" class="button">Buy Now</a>
    </div>
    <div class="card">
        
        <h3>Product 4</h3>
        <p> ₹2,000</p>
        <a href="#" class="button">Buy Now</a>
    </div>
</div>
</body>
</html>

```

OUTPUT:



ELECTRONICS CODE:

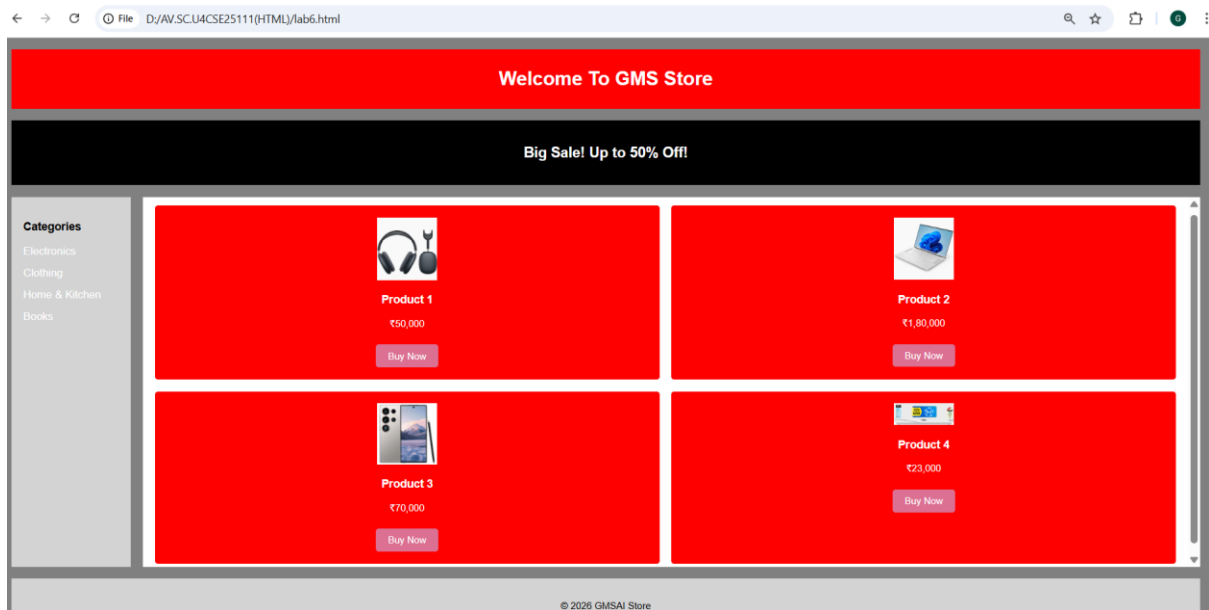
```
<!DOCTYPE html>
<html>
  <head>
    <style>
      body{
        font-family: Arial, sans-serif;
        margin: 0;
        padding: 0;
        background-color: grey;
      }
      .products{
        grid-area: products;
        background-color: white;
        padding: 20px;
        display: grid;
        grid-template-columns: repeat(2, 1fr);
        gap: 20px;
      }
      .card{
        background-color:red;
        color: white;
        padding: 20px;
        border-radius: 5px;
        text-align:center;
      }
      .button{
        display: inline-block;
        padding: 10px 20px;
        background-color: palevioletred;
        color: white;
        text-decoration: none;
        border-radius: 5px;
        margin-top: 10px;
      }
      .button:hover{
        background-color: plum;
      }
    </style>
  <body>
    </div>
    <div class="products">
      <div class="card">
        
```

```

        <h3>Product 1</h3>
        <p>₹50,000</p>
        <a href="#" class="button">Buy Now</a>
    </div>
    <div class="card">
        
        <h3>Product 2</h3>
        <p> ₹1,80,000</p>
        <a href="#" class="button">Buy Now</a>
    </div>
    <div class="card">
        
        <h3>Product 3</h3>
        <p> ₹70,000</p>
        <a href="#" class="button">Buy Now</a>
    </div>
    <div class="card">
        
        <h3>Product 4</h3>
        <p> ₹23,000</p>
        <a href="#" class="button">Buy Now</a>
    </div>
</div>
</body>
</html>

```


OUTPUT:



CLOTHING CODE:

CLOTHING CODE:

```
<!DOCTYPE html>

<html>
  <head>
    <style>
      body{
        font-family: Arial, sans-serif;
        margin: 0;
        padding: 0;
        background-color: grey;
      }
      .products{
        grid-area: products;
        background-color: white;
        padding: 20px;
        display: grid;
        grid-template-columns: repeat(2, 1fr);
        gap: 20px;
      }
      .card{
        background-color:red;
        color: white;
        padding: 20px;
        border-radius: 5px;
        text-align:center;
      }
      .button{
```

```

display: inline-block;
padding: 10px 20px;
background-color: palevioletred;
color: white;
text-decoration: none;
border-radius: 5px;
margin-top: 10px;
}
.button:hover{
background-color: plum;
}
</style>
<body>
<div>
<div class="products">
<div class="card">

<h3>Product 1</h3>
<p>₹400</p>
<a href="#" class="button">Buy Now</a>
</div>
<div class="card">

<h3>Product 2</h3>
<p>₹38000</p>
<a href="#" class="button">Buy Now</a>
</div>
<div class="card">

<h3>Product 3</h3>
<p>₹450</p>
<a href="#" class="button">Buy Now</a>
</div>
<div class="card">

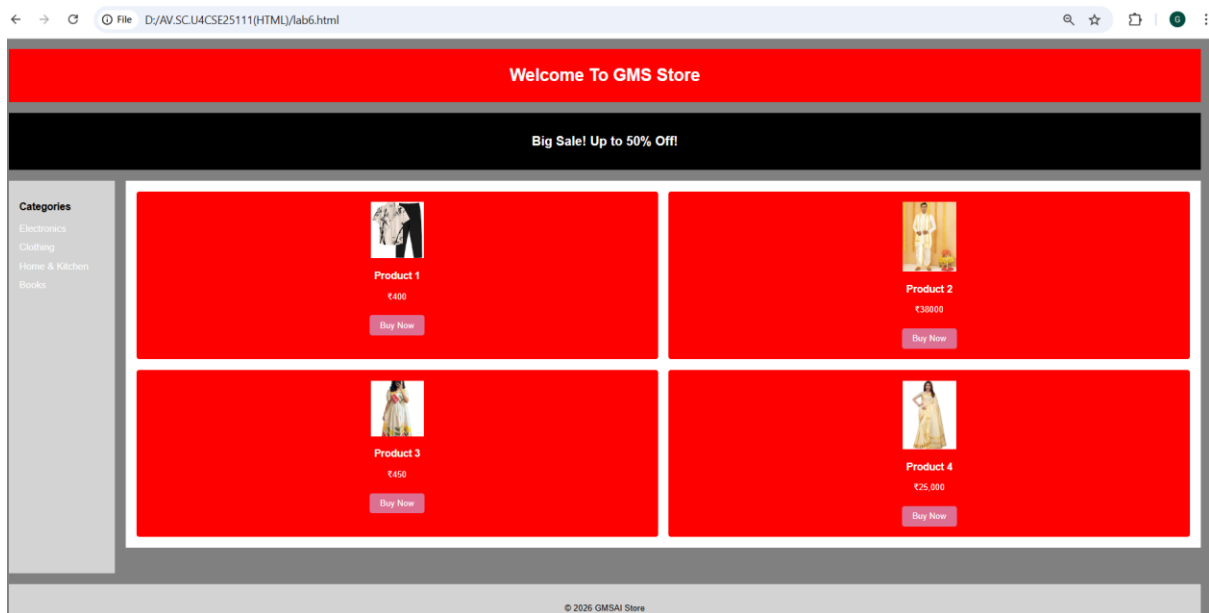

```

```

    <h3>Product 4</h3>
    <p> ₹25,000</p>
    <a href="#" class="button">Buy Now</a>
  </div>
</div>
</body>
</html>

```

OUTPUT:



BOOKS CODE:

```

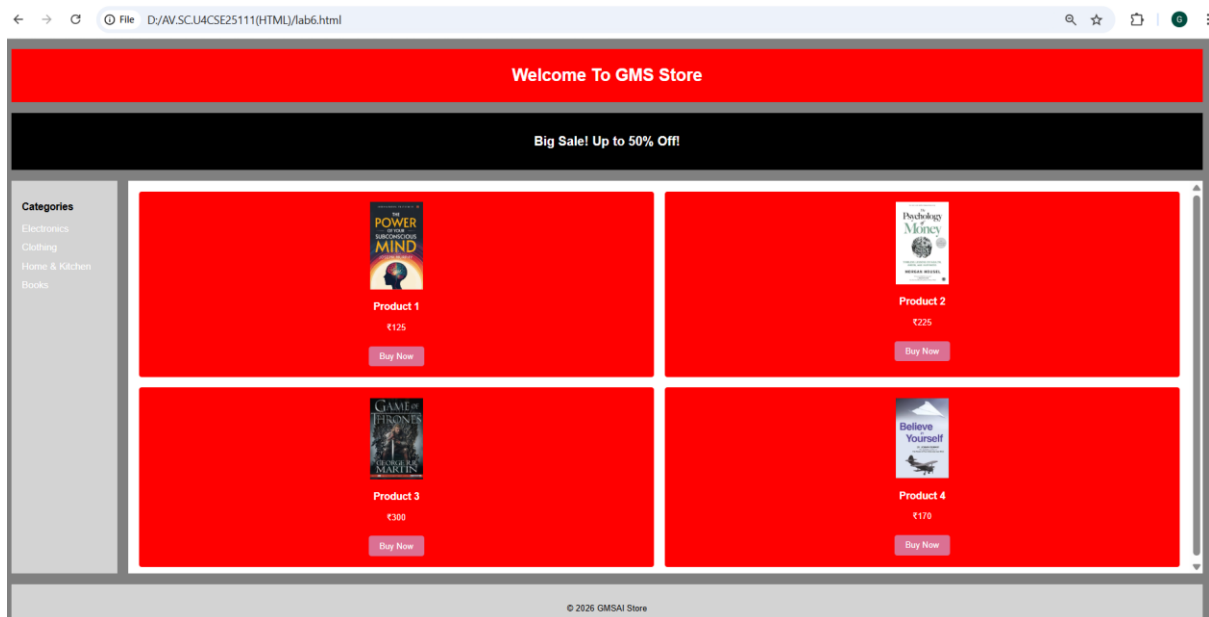
<!DOCTYPE html>
<html>
  <head>
    <style>
      body{
        font-family: Arial, sans-serif;
        margin: 0;
        padding: 0;
        background-color: grey;
      }
      .products{
        grid-area: products;
        background-color: white;
        padding: 20px;
        display: grid;
        grid-template-columns: repeat(2, 1fr);
        gap: 20px;
      }
      .card{
        background-color:red;

```

```

        color: white;
        padding: 20px;
        border-radius: 5px;
        text-align:center;
    }
    .button{
        display: inline-block;
        padding: 10px 20px;
        background-color: palevioletred;
        color: white;
        text-decoration: none;
        border-radius: 5px;
        margin-top: 10px;
    }
    .button:hover{
        background-color: plum;
    }
</style>
<body>
    </div>
    <div class="products">
        <div class="card">
            
            <h3>Product 1</h3>
            <p>₹125</p>
            <a href="#" class="button">Buy Now</a>
        </div>
        <div class="card">
            
            <h3>Product 2</h3>
            <p> ₹225</p>
            <a href="#" class="button">Buy Now</a>
        </div>
        <div class="card">
            
            <h3>Product 3</h3>
            <p> ₹300</p>
            <a href="#" class="button">Buy Now</a>
        </div>
        <div class="card">
            
    <h3>Product 4</h3>
    <p> ₹170</p>
    <a href="#" class="button">Buy Now</a>
</div>
</div>
</body>
</html>
```

OUTPUT:



EXPLANATION:

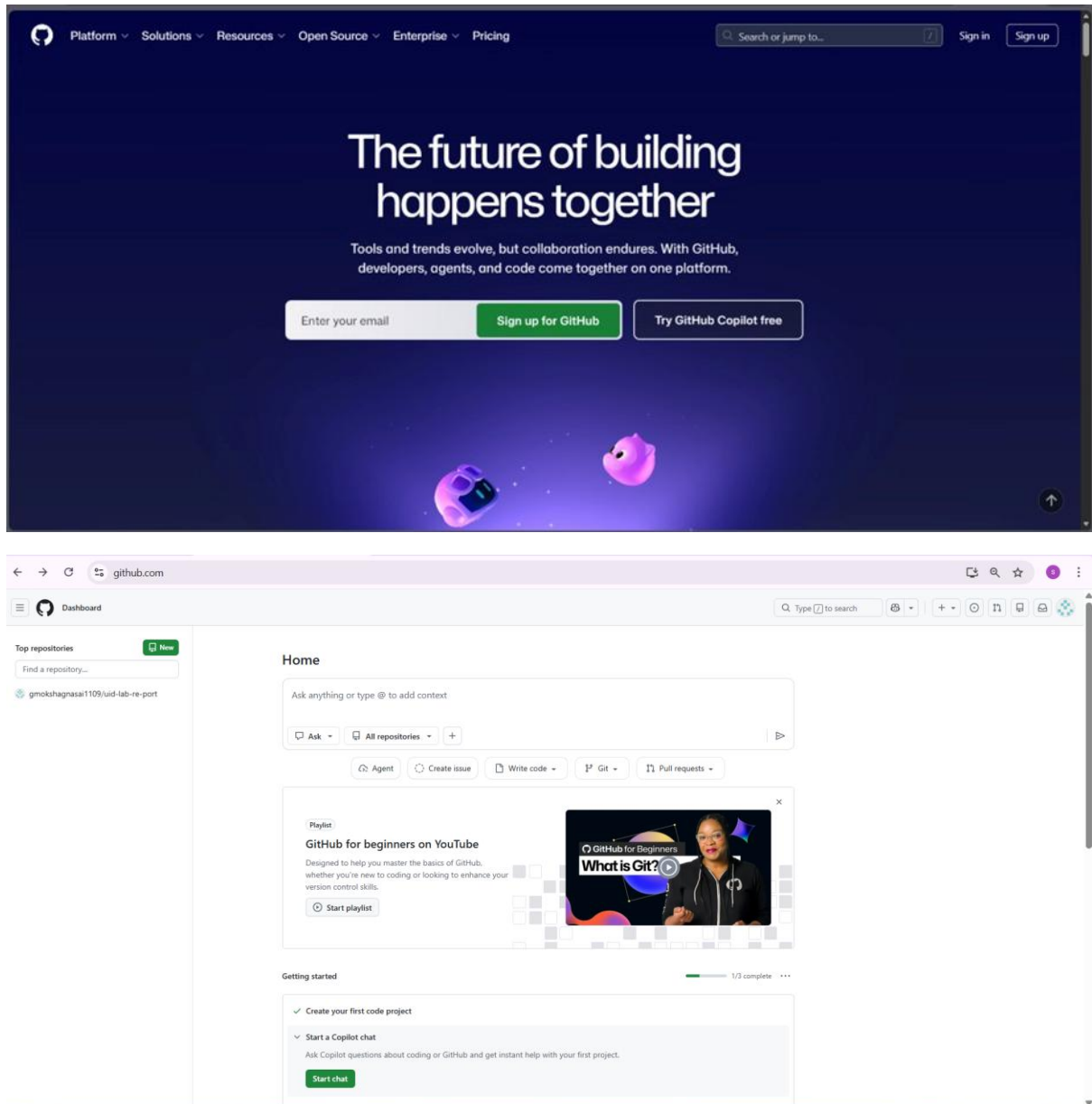
The HTML code creates an e-commerce webpage layout using CSS Grid and iframe-based navigation. The `<!DOCTYPE html>` declaration defines the HTML5 document type, and the `<html>` tag is the root element. Inside the `<head>`, the `<title>` sets the page name as “E-Commerce store,” and internal CSS is used to design the layout. The `<body>` contains a main `<div>` with class `.container` that uses `display: grid` to organize the page into sections like header, sale banner, sidebar, product area, and footer using `grid-template-areas`. The `.header` displays the store title, and the `.sale` section highlights promotional offers. The `.sidebar` contains navigation links using `<ul>` and `<li>`, where `<a>` tags with the `target="contentFrame"` attribute load different category pages (Electronics, Clothing, Home & Kitchen, Books) into the iframe. The `.products` section contains an `<iframe>` that acts as a dynamic display area for loading selected category pages. Styling like colors, padding, hover effects, and transitions improve the visual appearance. Finally, the `.footer` displays copyright information. Overall, the page combines HTML structure with CSS Grid and iframe to create a structured, interactive e-commerce layout.

LAB-7

Aim: To create the repository in GIT Hub and push or upload files of all the programs that are done to create different webpages and websites.

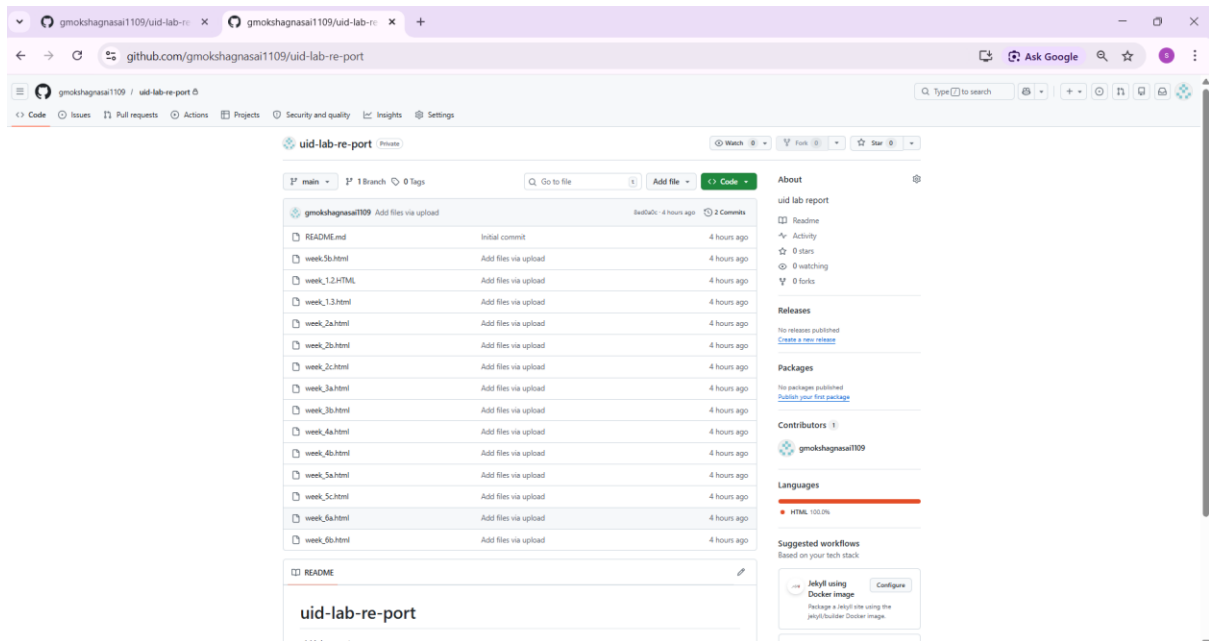
STEP-1:

Create a GitHub account Go to github.com, click Sign Up, and register using your email, username, and password.



STEP-2:

Create a new repository On GitHub, click New Repository, give it a name (e.g., Lab7-Webpages), choose public/private, and click Create and upload files.



EXPLANATION:

1. `git init` Initializes a new Git repository in your current folder. It creates a hidden `.git` directory where Git stores all version history.
2. `git add .` Stages all files in the current directory (the `.` means “everything here”). This tells Git which files you want to include in the next commit.
3. `git commit -m "First commit"` Saves a snapshot of the staged files into the repository with a message ("First commit"). The message should describe what you changed or added.
4. `git branch -M main` Renames the current branch to main. GitHub uses main as the default branch name now (instead of master).
5. `git remote add origin <PASTE_YOUR_URL_HERE>` Links your local repository to a remote repository on GitHub. Replace `<PASTE_YOUR_URL_HERE>` with the HTTPS or SSH URL of your GitHub repo.
6. `git push -u origin main` Pushes your local main branch to the remote repository named origin (GitHub). The `-u` sets up tracking so future git push commands know where to send changes.

REPOSITORY:

<https://github.com/gmokshagnasai1109/uid-lab-re-port>

LAB-8

8A.

AIM:

Create a webpage that displays a square that continuously rotates 360° using CSS animation.

PROGRAM:

MAIN CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Rotating Square</title>
  <style>
    body {
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      background-color: #f0f0f0;
    }

    .square {
      width: 100px;
      height: 100px;
      background-color: #3498db;
      animation: rotateSquare 2s linear infinite;
    }

    @keyframes rotateSquare {
      from {
        transform: rotate(0deg);
      }
      to {
        transform: rotate(360deg);
      }
    }
  </style>
</head>
<body>

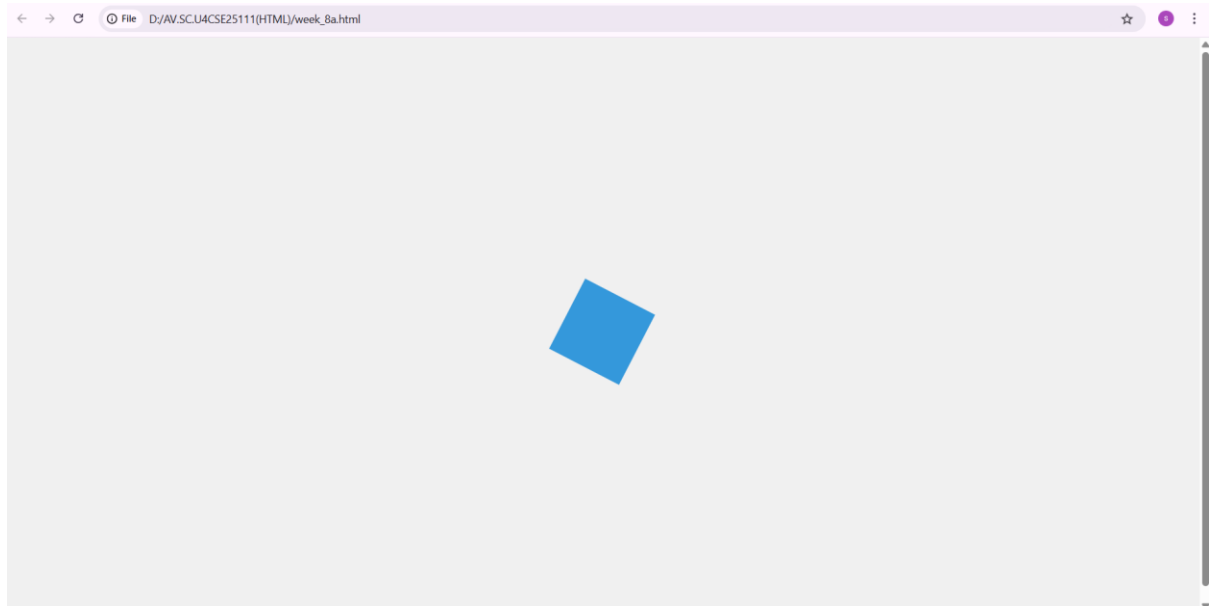
  <div class="square"></div>

</body>
```



```
</html>
```

OUTPUT:



EXPLANATION:

The HTML code creates a simple webpage that displays a continuously rotating square using CSS animation. The `<html>` structure includes a `<head>` section where the title is defined and internal CSS is written to style the page. The `<body>` is styled using Flexbox (`display: flex`) to center the square both horizontally and vertically on the screen, and `height: 100vh` ensures it takes the full viewport height. A `<div>` with the class `.square` is used to create the square shape by setting equal width and height and applying a background color. The animation is applied using the `animation` property, which links to `@keyframes rotateSquare`. Inside the keyframes, the `transform: rotate()` property rotates the square from 0 degrees to 360 degrees, creating a smooth continuous spinning effect. The `2s linear infinite` ensures the rotation is constant, smooth, and repeats endlessly.

8B.

AIM:

Create a ball that moves up and down continuously using CSS animation.

PROGRAM:

MAIN CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Bouncing Ball</title>
  <style>
    body {
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      background-color: #f0f0f0;
    }

    .ball {
      width: 80px;
      height: 80px;
      background-color: red;
      border-radius: 50%;
      animation: bounce 1s ease-in-out infinite;
    }

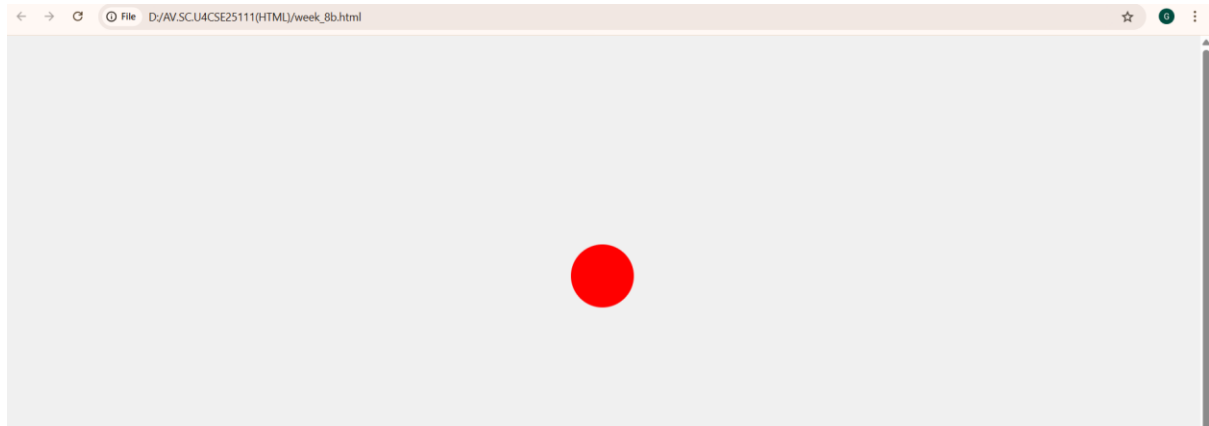
    @keyframes bounce {
      0% {
        transform: translateY(0);
      }
      50% {
        transform: translateY(-150px);
      }
      100% {
        transform: translateY(0);
      }
    }
  </style>
</head>
<body>

  <div class="ball"></div>

</body>
```

```
</html>
```

OUTPUT:



EXPLANATION:

The HTML code creates a webpage that displays a bouncing ball animation using CSS. The `<html>` structure includes a `<head>` section where the title is defined and styling is added using internal CSS. The `<body>` is designed using Flexbox (`display: flex`) to center the ball both horizontally and vertically on the screen, and `height: 100vh` ensures the content fills the full viewport height. A `<div>` with the class `.ball` is used to create the ball shape by setting equal width and height and applying `border-radius: 50%` to make it circular, along with a background color. The animation is applied using the `animation` property, which connects to the `@keyframes bounce`. In the keyframes, the `transform: translateY()` property moves the ball up and down—starting at its original position, moving upward at 50%, and returning back at 100%. The `1s ease-in-out infinite` makes the motion smooth and continuous, giving the effect of a ball bouncing repeatedly.

8C.

AIM:

Create a circle that moves horizontally and rotates at the same time.

PROGRAM:

MAIN CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Moving & Rotating Circle</title>
  <style>
    body {
      margin: 0;
      overflow: hidden;
      background-color: #f0f0f0;
    }

    .circle {
      width: 80px;
      height: 80px;
      background-color: purple;
      border-radius: 50%;
      position: absolute;
      top: 50%;
      left: 0;
      transform: translateY(-50%);
      animation: moveRotate 4s linear infinite;
    }

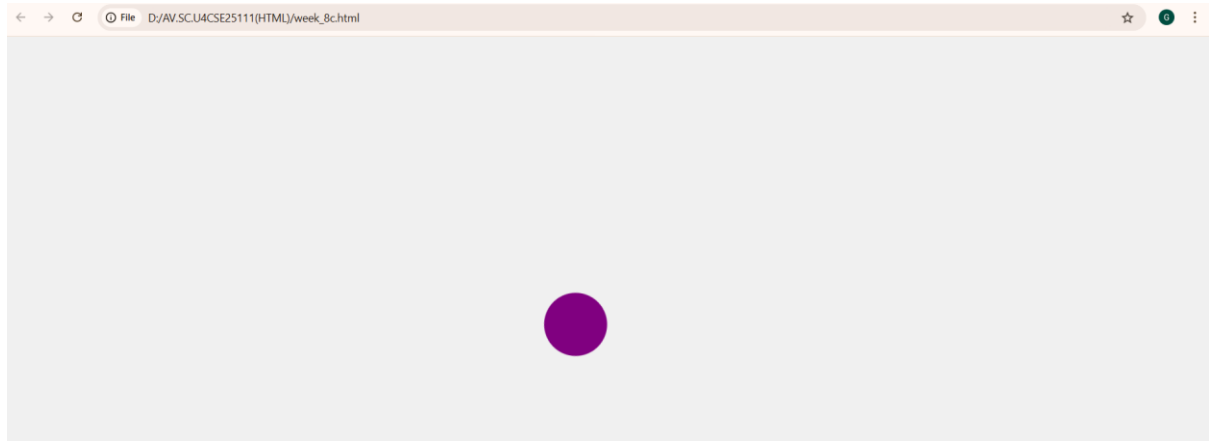
    @keyframes moveRotate {
      0% {
        left: 0;
        transform: translateY(-50%) rotate(0deg);
      }
      50% {
        left: 80%;
        transform: translateY(-50%) rotate(180deg);
      }
      100% {
        left: 0;
        transform: translateY(-50%) rotate(360deg);
      }
    }
  </style>
</head>
```

```
<body>

  <div class="circle"></div>

</body>
</html>
```

OUTPUT:



EXPLANATION:

The HTML code creates a webpage that shows a circle moving horizontally while rotating at the same time using CSS animation. The `<html>` structure includes a `<head>` section where the title and internal CSS styling are defined. The `<body>` has no margin and uses `overflow: hidden` to prevent scrollbars while the animation runs. A `<div>` with the class `.circle` is used to create the shape by setting equal width and height and applying `border-radius: 50%` to make it circular, along with a background color. The circle is positioned using `position: absolute`, starting from the left side and vertically centered using `top: 50%` and `transform: translateY(-50%)`. The animation is applied using the `animation` property, which links to `@keyframes moveRotate`. In the keyframes, the `left` property moves the circle from left to right and back, while `transform: rotate()` rotates it from 0° to 360° . The `4s linear infinite` makes the motion smooth and continuous, combining both movement and rotation effects together.

Lab-9

9-

Aim: Create an webpage using html, CSS, java script based on the given conditions:

- a) Create a button for each program, based on the button clicked the corresponding program has to be executed and should display the output.
- b) The logics for each button includes finding an even numbers, odd number, prime or not, factorial of a number, amstrong or not.
- c) Use appropriate designs to display the output.

Procedure:

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Checking a Number</title>
</head>
<body style="background-image:url(water.jpg);background-repeat: no-repeat; background-size: cover;">
  <table border="1" cellspacing="3" cellpadding="5"
    style="width: 600px; margin: 100px auto; padding:20px;
    background-color: rgba(255, 255, 255, 0.2);
    backdrop-filter: blur(10px);
    border-radius: 10px; color: black;">
    <tr>
      <td>To check even or odd:</td>
      <td><input type="number" id="num"></td>
      <td><button onclick="checkEvenOdd()">Check</button></td>
    </tr>
    <tr>
      <td>To check prime or not:</td>
      <td><input type="number" id="prime"></td>
      <td><button onclick="checkPrime()">Check</button></td>
    </tr>
    <tr>
      <td>To get factorial:</td>
      <td><input type="number" id="fact"></td>
      <td><button onclick="checkFactorial()">Check</button></td>
    </tr>
  </table>
```

```

        <td>To verivy Armstrong number or not:</td>
        <td><input type="number" id="arm"></td>
        <td><button onclick="checkArmstrong()">Check</button></td>
    </tr>
</table>

<script>
    function checkEvenOdd() {
        let num = Number(document.getElementById("num").value);
        if (num % 2 === 0) {
            alert("Even");
        } else {
            alert("Odd");
        }
    }

    function checkPrime() {
        let num = Number(document.getElementById("prime").value);
        let isPrime = true;

        if (num <= 1) {
            isPrime = false;
        } else {
            for (let i = 2; i <= Math.sqrt(num); i++) {
                if (num % i === 0) {
                    isPrime = false;
                    break;
                }
            }
        }

        if (isPrime) {
            alert("Prime Number");
        } else {
            alert("Not a Prime Number");
        }
    }

    function checkFactorial() {
        let num = Number(document.getElementById("fact").value);
        let fact = 1;

        if (num < 0) {
            alert("Factorial not defined for negative numbers");
            return;
        }

        for (let i = 1; i <= num; i++) {

```

```

        fact *= i;
    }

    alert("Factorial = " + fact);
}

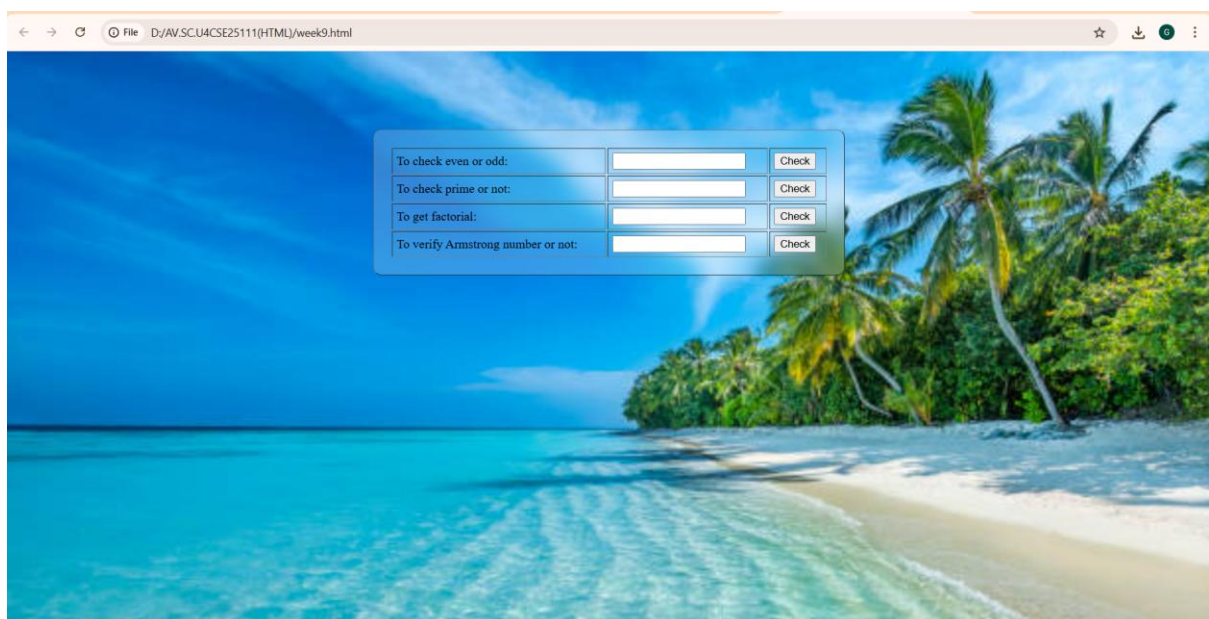
function checkArmstrong() {
    let num = Number(document.getElementById("arm").value);
    let temp = num;
    let sum = 0;
    let digits = num.toString().length;

    while (temp > 0) {
        let digit = temp % 10;
        sum += Math.pow(digit, digits);
        temp = Math.floor(temp / 10);
    }

    if (sum === num) {
        alert("Armstrong Number");
    } else {
        alert("Not an Armstrong Number");
    }
}
</script>
</body>
</html>

```

OUTPUT:



EXPLANATION:

The HTML code creates a webpage that allows users to perform different number-based checks using JavaScript functions. The `<html>` structure includes a `<head>` section with the title “Checking a Number,” and the `<body>` is styled with a background image that covers the entire screen. A `<table>` is used to organize the input fields and buttons neatly, with styling like padding, margin, semi-transparent background, and blur effect to improve appearance. Each row of the table contains a specific operation such as checking even or odd, prime number, factorial, and Armstrong number, where `<input>` fields are used to enter values and `<button>` elements trigger JavaScript functions using the `onclick` event. In the `<script>` section, functions like `checkEvenOdd()`, `checkPrime()`, `checkFactorial()`, and `checkArmstrong()` are defined to perform the respective logic. These functions take input values, process them using conditions and loops, and display the result using `alert()`. Overall, the code combines HTML for structure, CSS for styling, and JavaScript for functionality to create an interactive webpage for number operations.

Lab-10

10a-

Aim: write a Java script conditional statement to find the sign of product of four numbers using function. Display an alert box with specific sign

Procedure:

Code:

```
<!DOCTYPE html>
<html>{
<head>
  <title>Sign of Product</title>
</head>
<body>

<script>
  function findSign(a, b, c, d) {
    let product = a * b * c * d;
    let message;

    if (product > 0) {
      message = "The product is " + product + " and the sign is positive (+)";
    }
    else if (product < 0) {
      message = "The product is " + product + " and the sign is negative (-)";
    }
    else {
      message = "The product is " + product + " (zero)";
    }

    return message;
  }

  let num1 = Number(prompt("Enter 1st number:"));
  let num2 = Number(prompt("Enter 2nd number:"));
  let num3 = Number(prompt("Enter 3rd number:"));
  let num4 = Number(prompt("Enter 4th number:"));

  alert(findSign(num1, num2, num3, num4));
</script>

</body>
</html>
}
```

OUTPUT:

The first screenshot shows the prompt 'Enter 1st number:' with the input '12'. The second shows 'Enter 2nd number:' with '24'. The third shows 'Enter 3rd number:' with '36'. The fourth shows 'Enter 4th number:' with '48'. Each prompt is titled 'This page says' and has 'OK' and 'Cancel' buttons.

The final screenshot shows the result of the calculation: 'The product is 497664 and the sign is positive (+)'. The prompt is titled 'This page says' and has an 'OK' button.

EXPLANATION:

The HTML code creates a simple webpage that determines the sign of the product of four numbers using JavaScript. The `<html>` structure contains a `<head>` section with the title “Sign of Product,” and the `<body>` includes a `<script>` where all the logic is written. A function `findSign(a, b, c, d)` is defined to calculate the product of four numbers and determine whether the result is positive, negative, or zero using conditional statements (`if`, `else if`, `else`). The program takes user input using the `prompt()` function for four numbers, converts them into numeric values using `Number()`, and passes them to the function. The result is then displayed using the `alert()` function. Overall, the code demonstrates basic JavaScript concepts like functions, user input, arithmetic operations, and conditional statements to determine the sign of a product.

10b-

Aim: To Sort 3 number using functions. Display on alert box to show the result.

Procedure:

Code:

```
<!DOCTYPE html>
<html>{
<head>
  <title>Sort Three Numbers</title>
</head>
<body>

<script>
  function sortNumbers(a, b, c) {
    let result;

    if (a >= b && a >= c) {
      if (b >= c) {
        result = a + ", " + b + ", " + c;
      } else {
        result = a + ", " + c + ", " + b;
      }
    }
    else if (b >= a && b >= c) {
      if (a >= c) {
        result = b + ", " + a + ", " + c;
      } else {
        result = b + ", " + c + ", " + a;
      }
    }
    else {
      if (a >= b) {
        result = c + ", " + a + ", " + b;
      } else {
        result = c + ", " + b + ", " + a;
      }
    }

    return "Sorted numbers (descending): " + result;
  }

  let num1 = Number(prompt("Enter first number:"));
  let num2 = Number(prompt("Enter second number:"));
  let num3 = Number(prompt("Enter third number:"));
```

```
    alert(sortNumbers(num1, num2, num3));  
</script>  
  
</body>  
</html>  
}
```

OUTPUT:

The output shows three sequential browser windows for a web application titled "Sort Three Numbers".

- The first window prompts the user to "Enter first number:" with the input field containing "12". It has "OK" and "Cancel" buttons.
- The second window prompts the user to "Enter second number:" with the input field containing "14". It has "OK" and "Cancel" buttons.
- The third window prompts the user to "Enter third number:" with the input field containing "16". It has "OK" and "Cancel" buttons.

Below these, a larger screenshot shows the final result displayed in an alert box:

This page says
Sorted numbers (descending): 16, 14, 12

An "OK" button is visible at the bottom right of the alert box.

EXPLANATION:

This program is used to **sort three numbers in descending order (largest to smallest)**. It takes three inputs from the user using `prompt()` and converts them into numbers. Then, using **if-else conditions**, it compares the numbers to find which one is bigger and arranges them accordingly. Finally, it displays the sorted result using `alert()`. In short, it takes 3 numbers, compares them, and shows them from highest to lowest.

10c-

Aim: Write a JavaScript conditional statement to find the largest of five numbers using function and onclick event .Display an alert box to show the result.

Procedure:

Code:

```
<!DOCTYPE html>
<html>{
<head>
  <title>Largest of Four Numbers</title>
  <style>
    body {
      text-align: center;
      font-family: Arial;
      margin-top: 50px;
      background-color: orangered;
    }
    input {
      padding: 8px;
      margin: 5px;
      width: 150px;
      background-color: whitesmoke;
    }
    button {
      padding: 10px 20px;
      margin-top: 10px;
    }
  </style>
</head>
<body>

<h2>Find Largest of Five Numbers</h2>

<input type="number" id="1st" placeholder="Enter number 1"><br>
<input type="number" id="2nd" placeholder="Enter number 2"><br>
<input type="number" id="3rd" placeholder="Enter number 3"><br>
<input type="number" id="4th" placeholder="Enter number 4"><br>

<button onclick="findLargest()">Find Largest</button>

<script>
  function findLargest() {
```

```

let a = Number(document.getElementById("1st").value);
let b = Number(document.getElementById("2nd").value);
let c = Number(document.getElementById("3rd").value);
let d = Number(document.getElementById("4th").value);

let largest;

if (a >= b && a >= c && a >= d ) {
    largest = a;
}
else if (b >= a && b >= c && b >= d) {
    largest = b;
}
else if (c >= a && c >= b && c >= d) {
    largest = c;
}

else {
    largest = d;
}

alert("The largest number is: " + largest);
}
</script>

</body>
</html>
}

```

OUTPUT:



EXPLANATION:

This program is used to find the largest number among four inputs. It takes values from the user using input fields and converts them into numbers. Then, using if-else conditions, it compares all four numbers to determine which one is the greatest. Finally, it displays the

largest number using an alert().In short, it takes 4 numbers, compares them, and shows the biggest one.

10d-

Aim: To iterate the form 0 to given number, for each iteration to check if the current number is odd or even, and display the message to the screen.

Procedure:

Code:

```
<!DOCTYPE html>
<html>{
<head>
  <title>Odd or Even</title>
  <style>
    body {
      font-family: Arial;
      text-align: center;
      margin-top: 50px;
      background-color: salmon;
    }
    input, button {
      padding: 8px;
      margin: 5px;
      background-color: silver;
    }
  </style>
</head>
<body>

<h2>Odd or Even Numbers</h2>

<input type="number" id="limit" placeholder="Enter a number">
<br>
<button onclick="checkNumbers()">Check</button>

<div id="output"></div>

<script>
  function checkNumbers() {
    let n = Number(document.getElementById("limit").value);
    let result = "";
```



```

for (let i = 0; i <= n; i++) {
    if (i % 2 === 0) {
        result += i + " is Even <br>";
    } else {
        result += i + " is Odd <br>";
    }
}

document.getElementById("output").innerHTML = result;
}
</script>

</body>
</html>
}

```

OUTPUT:



EXPLANATION:

This program is used to check whether numbers are odd or even up to a given limit. It takes a number as input from the user. Then, using a for loop, it goes from 0 to that number and checks each value. If a number is divisible by 2, it is even; otherwise, it is odd. Finally, it displays the result on the webpage. In short, it takes a number and shows all odd and even numbers up to that limit.

10e-

Aim: To write a JS program to read 6 subject marks of a student and then compute the average marks of the student. Then the average is used to determine the corresponding grade.

Procedure:

Code:

```
<!DOCTYPE html>
<html> {
<head>
  <title>Student Grade Calculator</title>
  <style>
    body {

      font-family: Arial;
      text-align: left;
      margin-top: 50px;
      background-color: blueviolet;
    }
    input {
      padding: 8px;
      margin: 5px;
      width: 150px;
      background-color: greenyellow;
    }
    button {
      padding: 10px 20px;
    }
  </style>
</head>
<div>
  <style background-image: url(LOGO.png) ;background-repeat: no-repeat; width: 100px;
height: 100px;text-align: right;></style>
</div>
<body>

<h2>Grade Calculator</h2>
<h3>GGI</h3>
<input type="number" id="s1" placeholder="Enter Marks"><br>
```

```

<h3>UID</h3>
<input type="number" id="s2" placeholder="Enter Marks"><br>
<h3>OOPS</h3>
<input type="number" id="s3" placeholder="Enter Marks"><br>
<h3>Discrete Maths</h3>
<input type="number" id="s4" placeholder="Enter Marks"><br>
<h3>Modern Physics</h3>
<input type="number" id="s5" placeholder="Enter Marks"><br>
<h3>Linear Algebra</h3>
<input type="number" id="s6" placeholder="Enter Marks"><br>
<button onclick="calculateGrade()">Calculate</button>

<script>
function calculateGrade() {
    let m1 = Number(document.getElementById("s1").value);
    let m2 = Number(document.getElementById("s2").value);
    let m3 = Number(document.getElementById("s3").value);
    let m4 = Number(document.getElementById("s4").value);
    let m5 = Number(document.getElementById("s5").value);
    let m6 = Number(document.getElementById("s6").value);

    let marks = [m1, m2, m3, m4, m5, m6];

    // Validation: check empty, NaN, and range
    for (let i = 0; i < marks.length; i++) {
        if (isNaN(marks[i]) || marks[i] < 0 || marks[i] > 100) {
            alert("Please enter valid marks between 0 and 100 for all subjects!");
            return;
        }
    }

    let total = m1 + m2 + m3 + m4 + m5 + m6;
    let avg = total / 6;

    let grade;

    if (avg >= 90) {
        grade = "A+";
    } else if (avg >= 75) {
        grade = "A";
    } else if (avg >= 60) {
        grade = "B";
    } else if (avg >= 50) {
        grade = "C";
    } else if (avg >= 40) {
        grade = "D";
    } else {
        grade = "Fail";
    }
}

```

```
}  
  
    alert("Average Marks: " + avg.toFixed(2) + "\nGrade: " + grade);  
}  
</script>  
  
</body>  
</html>  
}
```

OUTPUT:

Student Grade Calculator

← → ↻ File D:/AV.SC.U4CSE251

Grade Calculator

GGI

UID

OOPS

Discrete Maths

Modern Physics

Linear Algebra

Calculate

Student Grade Calculator

D:/AV.SC.U4CSE25111(HTML)/week-10/10e.html

Grade Calculator

GGI

90

UID

90

OOPS

90

Discrete Maths

90

Modern Physics

90

Linear Algebra

90

Calculate

This page says

Average Marks: 90.00

Grade: A+

OK

EXPLANATION:

This program is used to **calculate a student's average marks and grade**. It takes marks of six subjects as input from the user. First, it checks whether all entered marks are valid (between 0 and 100). Then, it calculates the **total and average marks**. Based on the average, it uses **if-else conditions** to assign a grade (A+, A, B, etc.). Finally, it displays the average and grade using an alert(). In short, it takes marks, calculates average, and shows the grade.

LAB-11

11a-

Aim: A website requires users to enter a username. The system should display a message while typing if the username is less than 5 characters.

Procedure:

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Username Validator</title>
<style>
body{
  font-family:Arial;
  background:orange;
  display:flex;
  justify-content:center;
  align-items:center;
  height:100vh;
}
.box{
  background:white;
  padding:30px;
  border-radius:10px;
  text-align:center;
}
input{
  padding:10px;
  width:220px;
}
button{
  padding:10px 20px;
  background:orange;
  color:white;
  border:none;
  border-radius:5px;
}
</style>
</head>
<body>

<div class="box">
  <h2>Enter Username</h2>
  <input type="text" id="username" placeholder="Enter username">
```

```

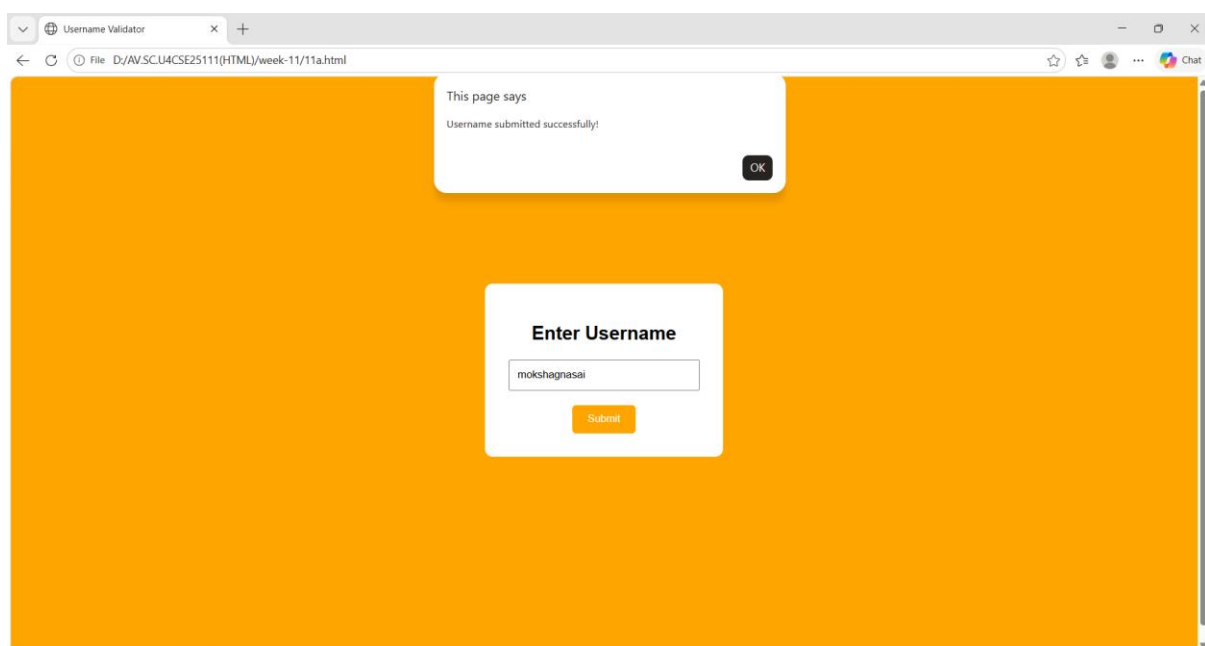
<br><br>
<button onclick="checkUsername()">Submit</button>
</div>

<script>
function checkUsername(){
  let username = document.getElementById("username").value;
  if(username.length < 5)
    alert("Username must be at least 5 characters long.");
  else
    alert("Username submitted successfully!");
}
</script>

</body>
</html>

```

OUTPUT:



EXPLANATION:

This HTML program creates a simple username validation webpage using HTML, CSS, and JavaScript. The `<!DOCTYPE html>` declaration defines the HTML5 document type, and the `<html>` tag is the root element containing the entire page. The `<head>` section includes the `<title>` tag to set the browser tab name as “Username Validator” and the `<style>` tag for internal CSS. The `<body>` is styled with properties like font-family, background, display: flex, justify-content, align-items, and height: 100vh to center the content on the screen. A `<div>` with class box is used to create a white container with padding, rounded corners, and centered text. The `<h2>` tag displays the heading, the `<input>` tag allows the user to enter a username,

and the <button> tag is used to submit the value by calling the checkUsername() function using the onclick attribute. In the <script> section, JavaScript gets the entered username using document.getElementById("username").value, checks its length using username.length, and displays an alert message. If the username has less than 5 characters, it shows an error message; otherwise, it shows a success message.

11b-

Aim: A registration form should validate email format before submission. If invalid, the form should not be submitted.

Procedure:

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Registration Form</title>
<style>
body{
  font-family:Arial;
  background:green;
  display:flex;
  justify-content:center;
  align-items:center;
  height:100vh;
  margin:0;
}
.box{
  background:white;
  padding:30px;
  border-radius:10px;
  text-align:center;
}
input{
  padding:10px;
  width:220px;
}
button{
  padding:10px 20px;
  background:green;
  color:white;
  border:none;
  border-radius:5px;
}
</style>
```

```

</head>
<body>

<div class="box">
  <h2>Register</h2>
  <input type="email" id="email" placeholder="Enter email">
  <br><br>
  <button onclick="checkEmail()">Register</button>
</div>

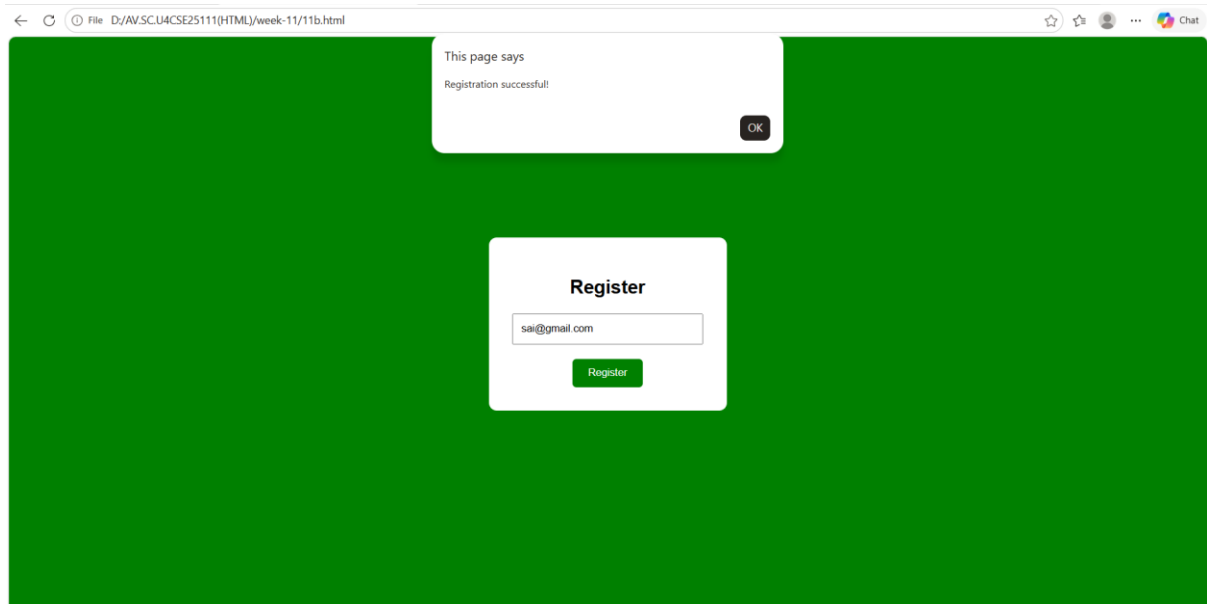
<script>
function checkEmail(){
  let email = document.getElementById("email").value;
  let emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

  if(!emailRegex.test(email))
    alert("Please enter a valid email address.");
  else
    alert("Registration successful!");
}
</script>

</body>
</html>

```

OUTPUT:



EXPLANATION:

This HTML program creates a simple registration form that validates an email address using HTML, CSS, and JavaScript. The `<!DOCTYPE html>` declaration defines the HTML5 document type, and the `<html>` tag is the root element of the webpage. The `<head>` section contains the `<title>` tag, which sets the browser tab name as “Registration Form,” and the `<style>` tag is used for internal CSS styling. The `<body>` is styled with properties like `font-family`, `background`, `display:flex`, `justify-content`, `align-items`, `height:100vh`, and `margin:0` to center the form on the page. A `<div>` with class `box` is used as the main container, and it contains a heading, an email input field, and a button. The `<input>` tag with `type="email"` allows the user to enter an email address, while the `<button>` uses the `onclick` attribute to call the `checkEmail()` function. In the `<script>` section, JavaScript gets the entered email using `document.getElementById("email").value`, checks it with a regular expression using `emailRegex.test(email)`, and displays an alert message. If the email format is wrong, it shows an error message; otherwise, it shows “Registration successful!”.

11c-

Aim: A system should check password strength while typing and display whether it is weak or strong.

Procedure:

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Password Checker</title>
<style>
body{
  font-family:Arial;
  background:linear-gradient(135deg, red, darkred);
  display:flex;
  justify-content:center;
  align-items:center;
  height:100vh;
  margin:0;
}
.box{
  background:white;
  padding:35px;
  border-radius:20px;
  text-align:center;
  width:300px;
  box-shadow:0 10px 25px rgba(0,0,0,0.3);
```

```

}
h2{
  color:darkred;
}
input{
  padding:12px;
  width:250px;
  border:2px solid red;
  border-radius:8px;
  outline:none;
}
#result{
  margin-top:15px;
  font-size:18px;
  font-weight:bold;
}
</style>
</head>
<body>

<div class="box">
  <h2>Password Checker</h2>
  <input type="password" id="password" placeholder="Enter password"
oninput="checkPassword()">
  <p id="result"></p>
</div>

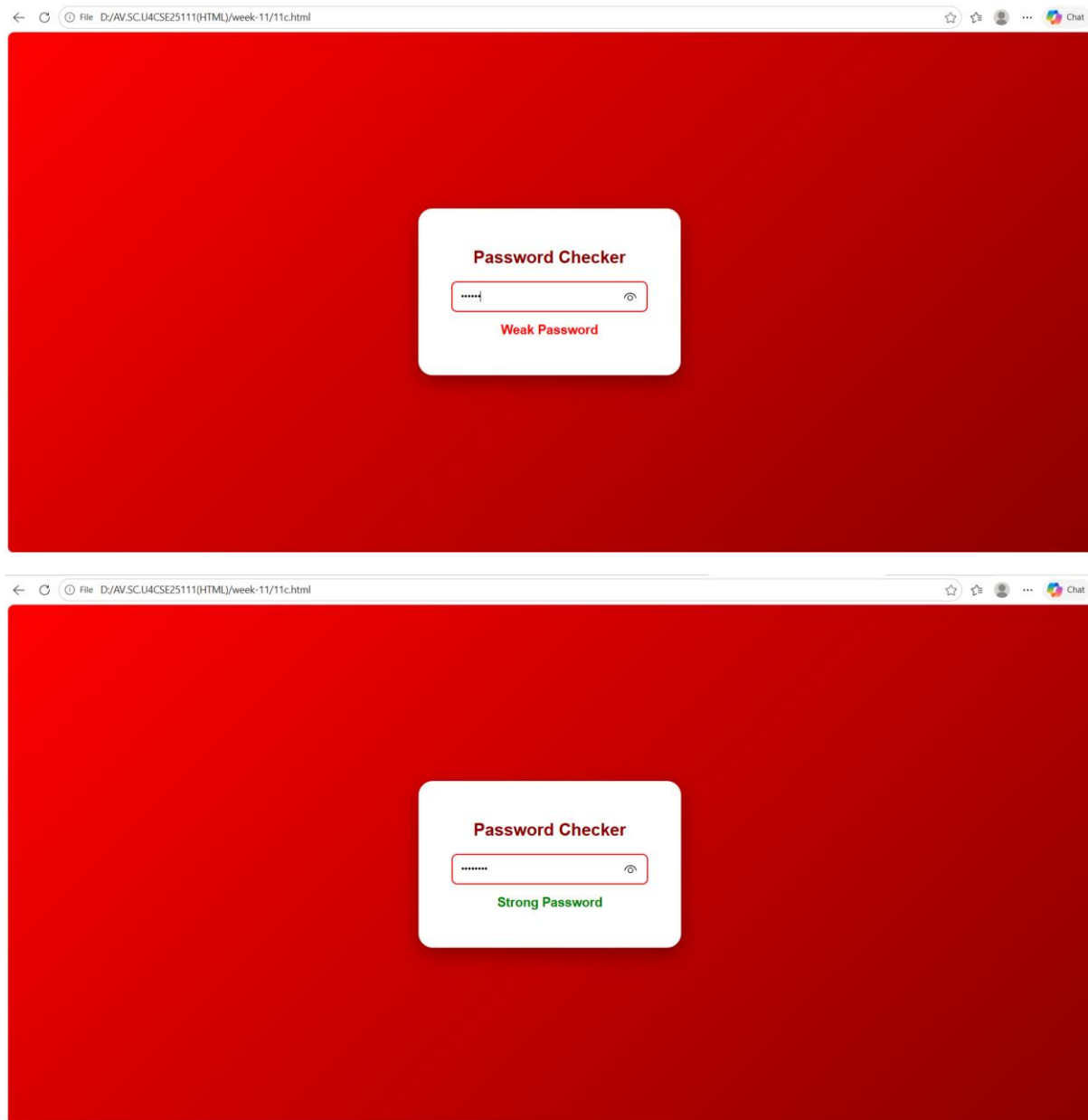
<script>
function checkPassword(){
  let password = document.getElementById("password").value;
  let result = document.getElementById("result");

  if(password.length == 0){
    result.innerHTML = "";
  }
  else if(password.length < 8){
    result.innerHTML = "Weak Password";
    result.style.color = "red";
  }
  else{
    result.innerHTML = "Strong Password";
    result.style.color = "green";
  }
}
</script>

</body>
</html>

```

OUTPUT:



EXPLANATION:

This HTML program creates a password strength checker using HTML, CSS, and JavaScript. The `<!DOCTYPE html>` declaration defines the HTML5 document type, and the `<html>` tag is the root element of the webpage. The `<head>` section contains the `<title>` tag and internal CSS using the `<style>` tag. The `<body>` is styled with a red gradient background and Flexbox properties like `display:flex`, `justify-content:center`, and `align-items:center` to place the box in the center of the page. A `<div>` with class `box` is used as the main container with padding, rounded corners, width, and shadow effect. The `<input>` tag with `type="password"` allows the user to enter a hidden password, and the `oninput` attribute calls the `checkPassword()` function whenever the user types. The `<p>` tag with `id result` is used to display the password strength message. In the `<script>` section, JavaScript gets the password value using `document.getElementById("password").value` and checks its length. If the password is empty,

no message is shown; if it is less than 8 characters, it displays “Weak Password” in red; otherwise, it displays “Strong Password” in green.

11d-

Aim: A form should validate a mobile number when the user leaves the input field. The number must be exactly 10 digits.

Procedure:

Code:

```
<!DOCTYPE html>
<html>
<head>
<title>Mobile Number Validation</title>
<style>
body{
  font-family:Arial;
  background:linear-gradient(135deg, blue, purple);
  display:flex;
  justify-content:center;
  align-items:center;
  height:100vh;
  margin:0;
}
.box{
  background:white;
  padding:30px;
  border-radius:15px;
  text-align:center;
  width:300px;
  box-shadow:0 10px 25px rgba(0,0,0,0.3);
}
input{
  padding:10px;
  width:230px;
  border:2px solid blue;
  border-radius:8px;
  outline:none;
}
#res{
  font-weight:bold;
  margin-top:15px;
}
</style>
```

```

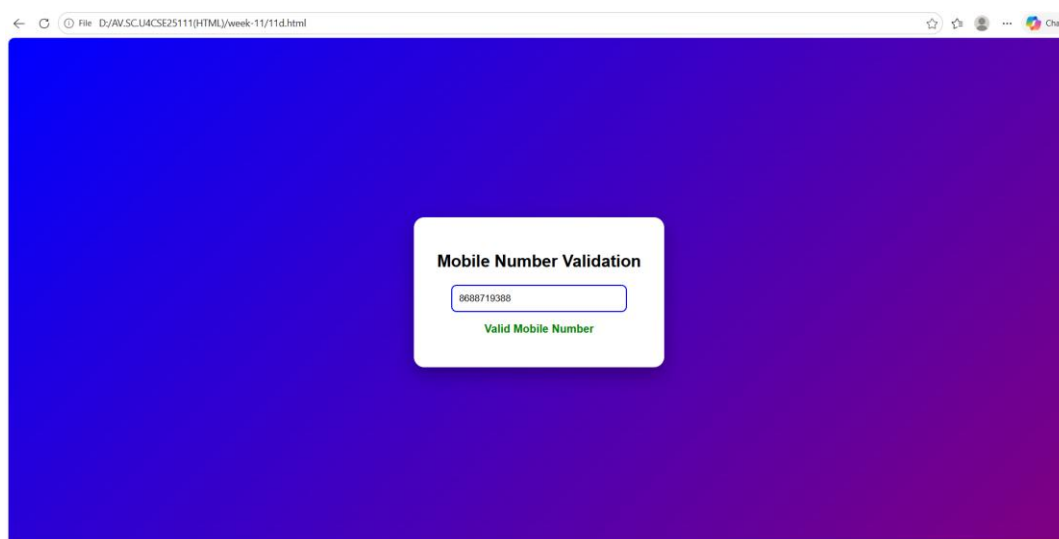
<script>
function validatenumner(){
  let x = document.getElementById("num").value;
  let pattern = /^[0-9]{10}$/;

  if(x == ""){
    document.getElementById("res").innerHTML = "";
  }
  else if(pattern.test(x)){
    document.getElementById("res").innerHTML = "Valid Mobile Number";
    document.getElementById("res").style.color = "green";
  }
  else{
    document.getElementById("res").innerHTML = "Invalid - Mobile number should have 10 digits";
    document.getElementById("res").style.color = "red";
  }
}
</script>
</head>

<body>
<div class="box">
  <h2>Mobile Number Validation</h2>
  <input type="text" id="num" placeholder="Enter mobile number"
oninput="validatenumner()">
  <p id="res"></p>
</div>
</body>
</html>

```

OUTPUT:



EXPLANATION:

This HTML program creates a mobile number validation webpage using HTML, CSS, and JavaScript. The `<!DOCTYPE html>` declaration defines the HTML5 document type, and the `<html>` tag is the root element of the webpage. The `<head>` section contains the `<title>` tag, internal CSS using the `<style>` tag, and JavaScript using the `<script>` tag. The `<body>` is styled with a blue and purple gradient background and Flexbox properties like `display: flex`, `justify-content: center`, and `align-items: center` to place the form box in the center of the page. A `<div>` with class `box` is used as the main container with padding, rounded corners, width, and shadow effect. The `<input>` tag allows the user to enter a mobile number, and the `oninput` attribute calls the `validatenumber()` function whenever the user types. In JavaScript, the entered value is taken using `document.getElementById("num").value`, and a regular expression `/^[0-9]{10}$/` is used to check whether the number contains exactly 10 digits. If the field is empty, no message is shown; if the number is valid, it displays “Valid Mobile Number” in green; otherwise, it displays an invalid message in red using the `<p>` tag with id `res`.

11e-

Aim: 5. Design an HTML page, such that the registration form must validate:

Name (not empty)

Email (valid format)

Password (minimum 6 characters)

Procedure:

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Registration Form Validation</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background: linear-gradient(135deg, #4facfe, #00f2fe);
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      margin: 0;
    }
  </style>
</head>
<body>
  <div class="box">
    <input type="text" id="name" />
    <input type="text" id="email" />
    <input type="password" id="password" />
    <button type="button" value="Register" />
  </div>
  <p id="res"></p>
</body>
</html>
```



```
.container {  
  background: white;  
  width: 350px;  
  padding: 30px;  
  border-radius: 15px;  
  text-align: center;  
  box-shadow: 0 10px 25px rgba(0, 0, 0, 0.25);  
}
```

```
h2 {  
  margin-bottom: 20px;  
  color: #333;  
}
```

```
label {  
  display: block;  
  text-align: left;  
  margin-top: 12px;  
  font-weight: bold;  
}
```

```
input {  
  width: 100%;  
  padding: 10px;  
  margin-top: 6px;  
  border: 2px solid #ccc;  
  border-radius: 8px;  
  box-sizing: border-box;  
  font-size: 15px;  
  outline: none;  
}
```

```
input:focus {  
  border-color: #4facfe;  
}
```

```
input[type="submit"] {  
  margin-top: 20px;  
  background: #4facfe;  
  color: white;  
  border: none;  
  cursor: pointer;  
  font-size: 16px;  
  font-weight: bold;  
}
```

```
input[type="submit"]:hover {
```

```

        background: #0078d7;
    }

    .message {
        margin-top: 15px;
        font-weight: bold;
        min-height: 20px;
    }

    .error {
        color: red;
    }

    .success {
        color: green;
    }
</style>
</head>

<body>

    <div class="container">
        <h2>Registration Form</h2>

        <form onsubmit="return validateForm()">
            <label>Name</label>
            <input type="text" id="name" placeholder="Enter your name"
oninput="clearMessage()">

            <label>Email</label>
            <input type="text" id="email" placeholder="Enter your email"
oninput="clearMessage()">

            <label>Password</label>
            <input type="password" id="password" placeholder="Enter password"
oninput="clearMessage()">

            <input type="submit" value="Register">
        </form>

        <p id="message" class="message"></p>
    </div>

    <script>
        function validateForm() {
            let name = document.getElementById("name").value.trim();
            let email = document.getElementById("email").value.trim();
            let password = document.getElementById("password").value;

```

```

let message = document.getElementById("message");

let emailPattern = /^[a-zA-Z0-9._-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;

if (name === "") {
    message.innerHTML = "Wrong: Name must not be empty.";
    message.className = "message error";
    return false;
}

if (!emailPattern.test(email)) {
    message.innerHTML = "Wrong: Please enter a valid email address.";
    message.className = "message error";
    return false;
}

if (password.length < 6) {
    message.innerHTML = "Wrong: Password must contain at least 6 characters.";
    message.className = "message error";
    return false;
}

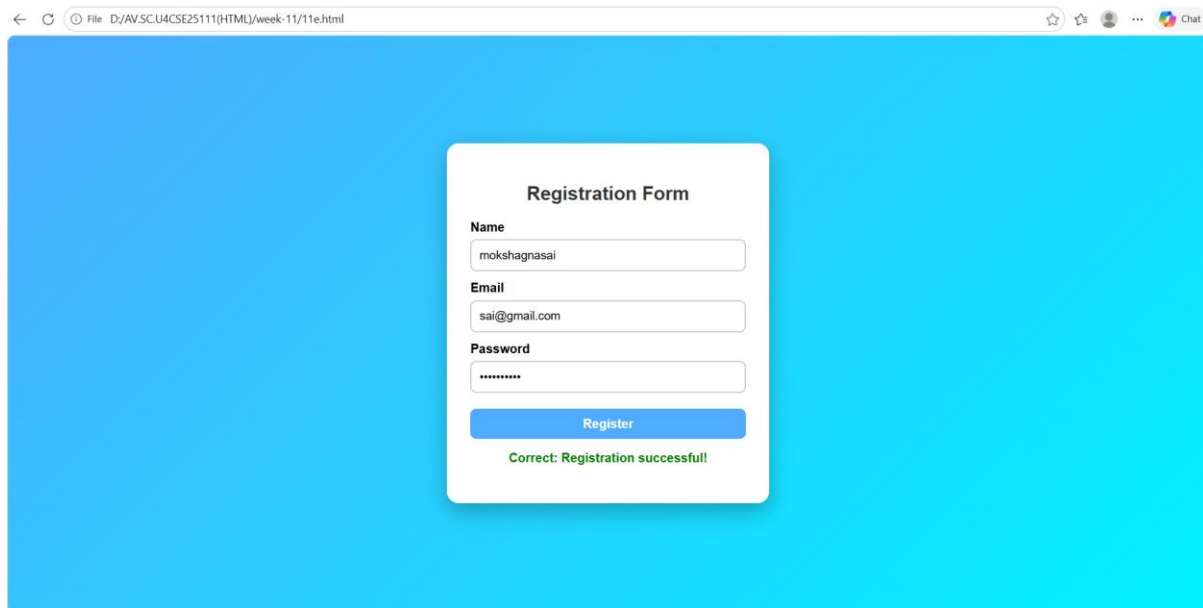
message.innerHTML = "Correct: Registration successful!";
message.className = "message success";
return false;
}

function clearMessage() {
    let message = document.getElementById("message");
    message.innerHTML = "";
    message.className = "message";
}
</script>

</body>
</html>

```

OUTPUT:



The screenshot shows a web browser window with a blue gradient background. In the center is a white registration form titled "Registration Form". The form contains three input fields: "Name" with the value "mokshagnasai", "Email" with the value "sai@gmail.com", and "Password" which is masked with dots. Below these fields is a blue "Register" button. Underneath the button, a green message states "Correct: Registration successful!". The browser's address bar at the top shows the file path "D:/AV/SC/U4CSE25111(HTML)/week-11/11e.html".

EXPLANATION:

This HTML program creates a registration form validation webpage using HTML, CSS, and JavaScript. The `<!DOCTYPE html>` declaration defines the HTML5 document type, and the `<html>` tag is the root element of the webpage. The `<head>` section contains the `<title>` tag and internal CSS using the `<style>` tag. The `<body>` is styled with a gradient background and Flexbox properties like `display:flex`, `justify-content:center`, and `align-items:center` to center the form. A `<div>` with class `container` is used as the main box, containing a heading and a `<form>`. The form uses `<label>` tags for field names and `<input>` tags for name, email, password, and submit button. The `onsubmit` attribute calls the `validateForm()` function, while `oninput` clears old messages when the user types. In JavaScript, values are taken using `document.getElementById()`, and `trim()` removes extra spaces. The email is checked using a regular expression, the name is checked to ensure it is not empty, and the password is checked to have at least 6 characters. If any value is wrong, an error message is shown in red; otherwise, a success message is shown in green using the `<p>` tag with id `message`.

LAB-12

Aim: To install and setup the Electron Framework application.

Procedure:

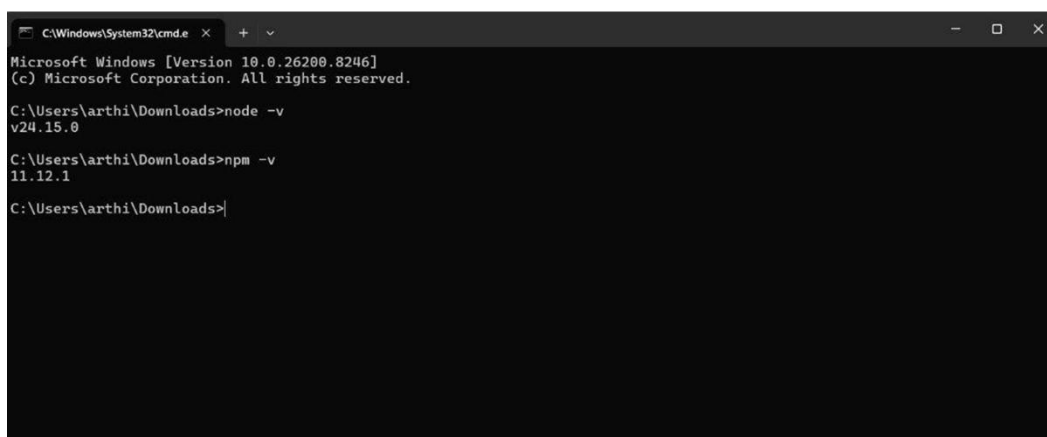
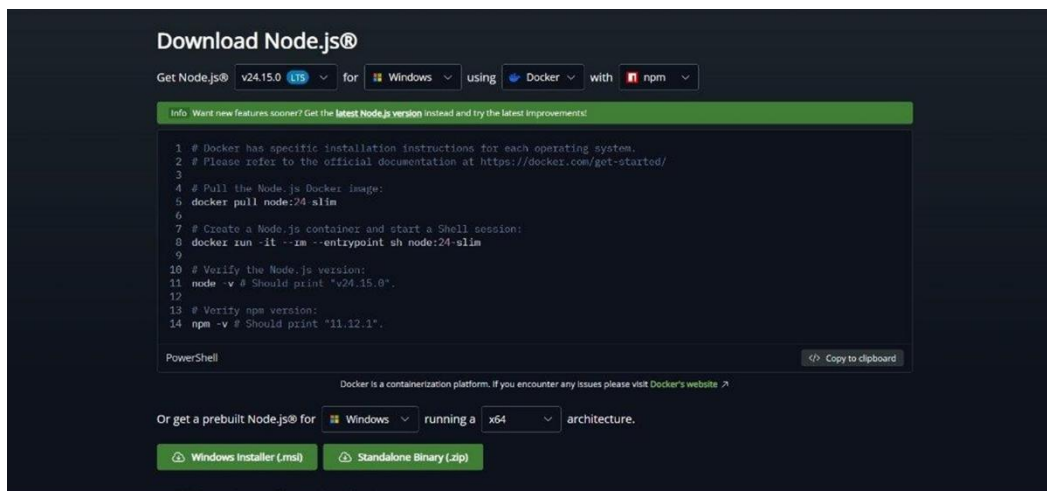
Step 1: Go to the web browser and download node.js.

Step 2: After installing the Node.js, complete the setup process.

Step 3: After installing node.js, Open the command prompt and navigate to the folder where your project is located.

Step 4: And give the prompt "node -v" and also "npm -v".

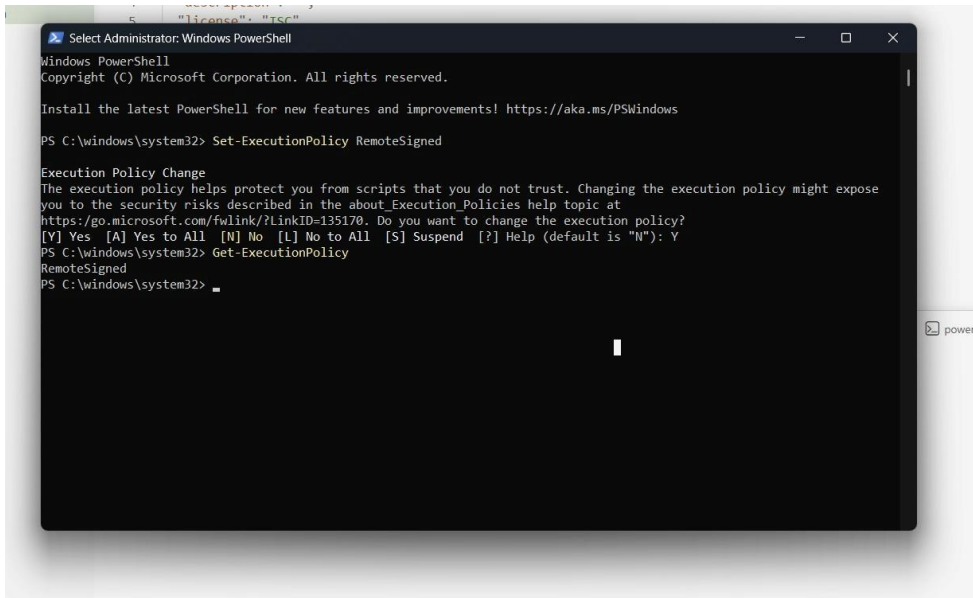
Step 5: If you get the versions then it is successfully installed.



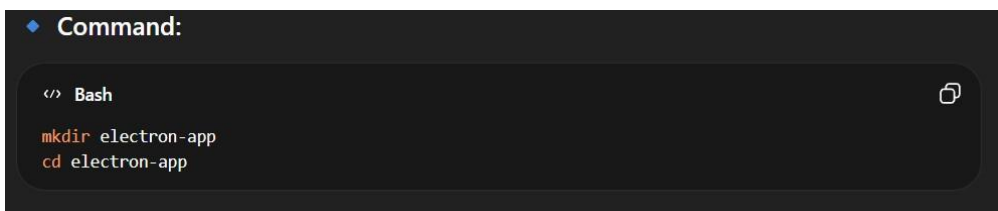
Step 6: Now open PowerShell as Administrator and give the prompt "Set-ExecutionPolicy RemoteSigned". 138

- This command allows local PowerShell scripts to run securely.

- And press Y.



Create Project Folder :



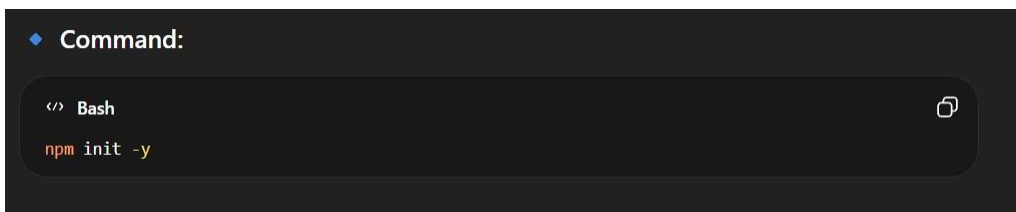
Step 7: Create a new folder for the Electron project using mkdir.

Step 8: Move into the project folder using cd.

- This helps keep all project files organized.

Step 9:

Initialize Node Project:



- The above command creates the package.json file.
- This file stores:
 1. Project name
 2. Version
 3. Main entry file

4. Scripts

3. Dependencies

- Without package.json file, npm cannot manage project properly.

Step 10:

Install Electron:

```
◆ Command:

</> Bash
npm install electron --save-dev
```

- Downloads Electron package
- Saves it in node_modules
- Adds it to devDependencies
- --save-dev→Because Electron is needed during development.

Step 11:

Configure package.json:

```
◆ Add:

</> JSON
{
  "main": "main.js",
  "scripts": {
    "start": "electron ."
  }
}
```

Main→Tells Electron which file starts the application.

Start→Defines command used when we run:

```
</> Bash
npm start
```

Electron → This command runs Electron in the current project folder.

Create Main Application Files:

```
◆ Required Files:

main.js
index.html
style.css
renderer.js
```

Step 12: main.js→Controls app lifecycle and creates windows

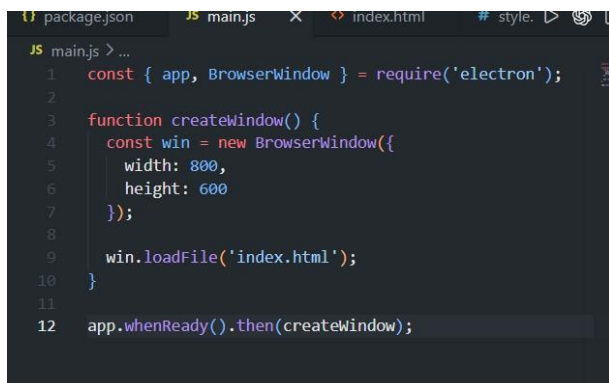
Step 13: index.html→User interface

Step 14: style.css→Design and layout

Step 15: renderer.js→Handles user interaction

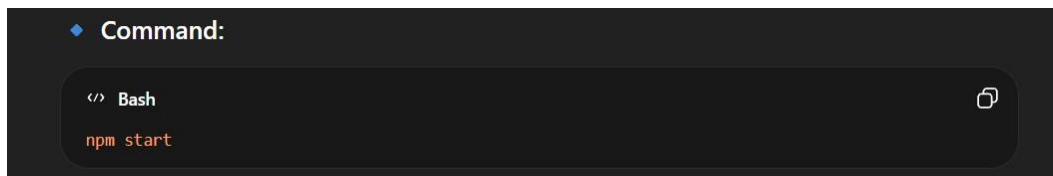
Write main.js

- Imports Electron modules
- Creates application window
- Loads UI



```
1  const { app, BrowserWindow } = require('electron');
2
3  function createWindow() {
4    const win = new BrowserWindow({
5      width: 800,
6      height: 600
7    });
8
9    win.loadFile('index.html');
10 }
11
12 app.whenReady().then(createWindow);
```

Run Application:



```
Command:
</ Bash
npm start
```